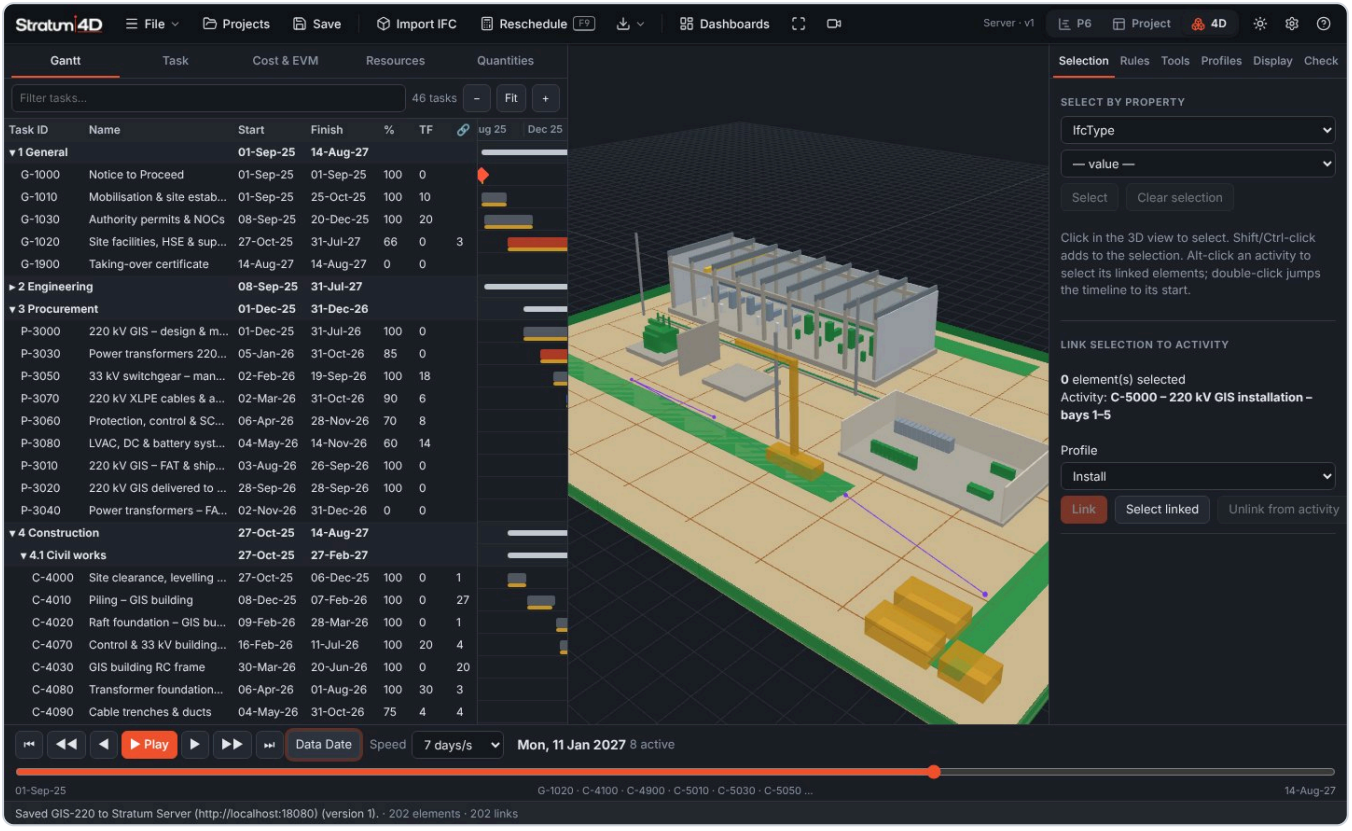
User Guide

Release 0.3 · Interface layouts, 4D simulation, project database, Primavera P6 connections, dashboards and emailed reports



Worked example: 220/33 kV GIS Substation, Abu Dhabi (fictional sample project)

Developed by **Junaid Quamar**, Senior Planning Engineer
linkedin.com/in/junaidquamar

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Contents

1. About this guide

2. Installing Stratum

3. First start

4. The workspace and the three layouts

5. Working with the schedule

6. 4D simulation

7. Cost, resources and quantities

8. The project database

9. Primavera P6 connections

10. Dashboards and reports

11. Stratum Server for teams

12. Keyboard shortcuts

13. Troubleshooting

14. Known limits

Appendix A · Example project activities

Appendix B · Terms in each layout

1. About this guide

This guide covers Stratum 4D release 0.3: the three interface layouts, scheduling, 4D simulation, the project database, Primavera P6 connections, and the dashboard and report module with emailed reports. Every screenshot comes from the application itself, running as a Stratum Server, with one example project loaded throughout.

The example project

The example is a **220/33 kV GIS substation in Abu Dhabi**, delivered as an EPC contract. It ships with Stratum as a sample (*Load sample → 220 kV GIS substation (UAE)*), so you can follow every step yourself. The project is fictional. Names, quantities and values are illustrative, and the Islamic public holidays in its calendar are approximate because they follow the moon.

Item	Example project
Project	GIS-220 · 220/33 kV GIS Substation, Abu Dhabi (sample project)
Scope	220 kV GIS hall with 9 bays and 10 t EOT crane; two 220/33 kV power transformers with firewall and oil pits; control & 33 kV building (33 kV switchgear, protection, control & SCADA, LVAC, DC); cable trenches and 220 kV XLPE cables; earthing grid and lightning masts; roads, drainage, boundary wall and gatehouse
Contract structure (WBS level 1)	General · Engineering · Procurement · Construction (Civil works, Electromechanical works, Testing & commissioning)
Schedule	46 activities, 62 relationships (FS, SS and FF with lags), activity codes *Discipline* and *Area*, user field *Package*
Calendar	UAE 6-day week, Monday–Saturday, 07:00–16:00 with a one-hour break; UAE public holidays 2025–2027
Resources and cost	15 resources in AED: labour (design engineers, civil crew, electrical technicians, OEM specialists, testing engineers), plant (200 t mobile crane, piling rig) and procured equipment (GIS, transformers, switchgear, panels, cables)
Data date	19-Oct-26
3D model	202 elements carrying the property `Pset_Stratum.ActivityID`, so one rule links the whole model

Status at the data date

Engineering is complete apart from the as-built documentation. The 220 kV GIS arrived on site on 28-Sep-26, three weeks after its baseline date, and installation of bays 1–5 has started. The power transformers are 85% through manufacture and running 21 days late, which puts transformer erection, pre-commissioning and energisation on the critical path. The forecast taking-over date is 14-Aug-27 against a baseline of 31-Jul-27.

Milestone	Name	Current / forecast	Baseline	Status
G-1000	Notice to Proceed	01-Sep-25	01-Sep-25	Achieved
P-3020	220 kV GIS delivered to site	28-Sep-26	07-Sep-26	Achieved

Milestone	Name	Current / forecast	Baseline	Status
C-6010	220 kV energisation	31-May-27	17-May-27	Forecast

> Critical path after rescheduling (F9): `G-1020` → `G-1900` → `P-3030` → `P-3040` → `C-4900` → `C-5030` → `C-5040` → `C-6000` → `C-6010` → `C-6020`. CPM finish: 12-Aug-27.

2. Installing Stratum

Stratum runs three ways. The engine and the screens are the same in all of them; what changes is where projects are stored and which services are available.

Way to run	Where projects live	P6 connections	Emailed reports	Typical use
Desktop app (Windows, macOS, Linux)	A local SQLite database in your user folder	Yes	Yes, while the app is open	One planner, offline work
Browser	The browser's own storage	No (use XER files)	No	Quick look, demos
Stratum Server (shared)	One PostgreSQL database for the team	Yes	Yes, around the clock	Project controls team

Desktop installers

Installers are built for every release: an NSIS installer for Windows, DMG files for Intel and Apple Silicon Macs, and AppImage/deb packages for Linux. They are produced by the project's GitHub Actions workflow and appear under *Actions → Desktop installers → latest run → Artifacts*.

The installers are not code-signed yet. On Windows, SmartScreen shows "Windows protected your PC": choose *More info → Run anyway*. On a Mac, right-click the app and choose *Open* the first time. The Windows installer lets you choose the install folder and adds Start-menu and desktop shortcuts. Double-clicking an `.xer` or `.ifc` file opens it in Stratum.

Stratum Server

See Stratum Server for teams for the Docker Compose setup, environment settings, users and roles.

3. First start

The first time Stratum opens, a short setup asks two questions. Both can be changed later in Settings.

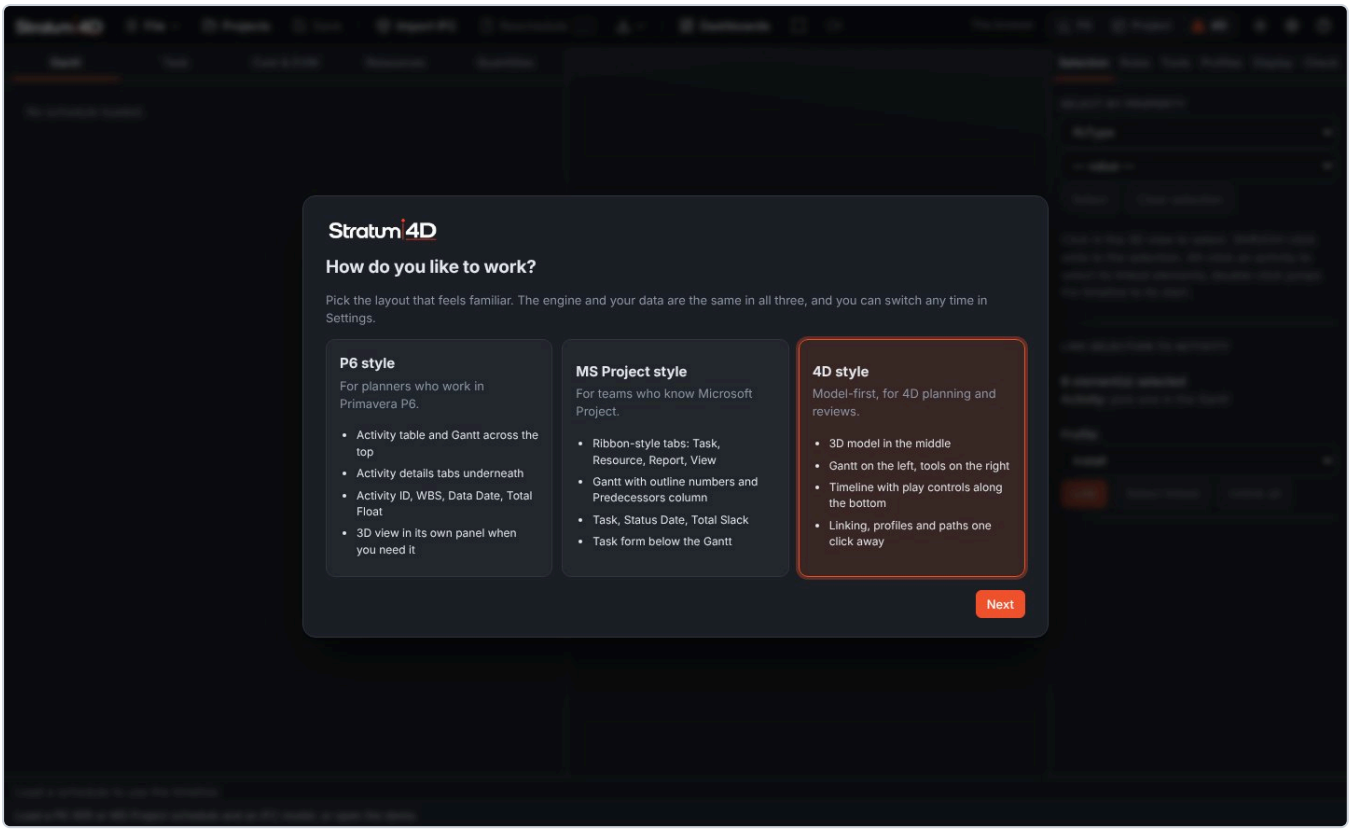


Figure 1. Step 1: choose the layout that feels familiar. The data and every feature are the same in all three.

Layout. *P6 style* suits planners who work in Primavera P6, *MS Project style* suits teams used to Microsoft Project, and *4D style* puts the 3D model in the middle for 4D planning and reviews. Section 4 describes each layout.

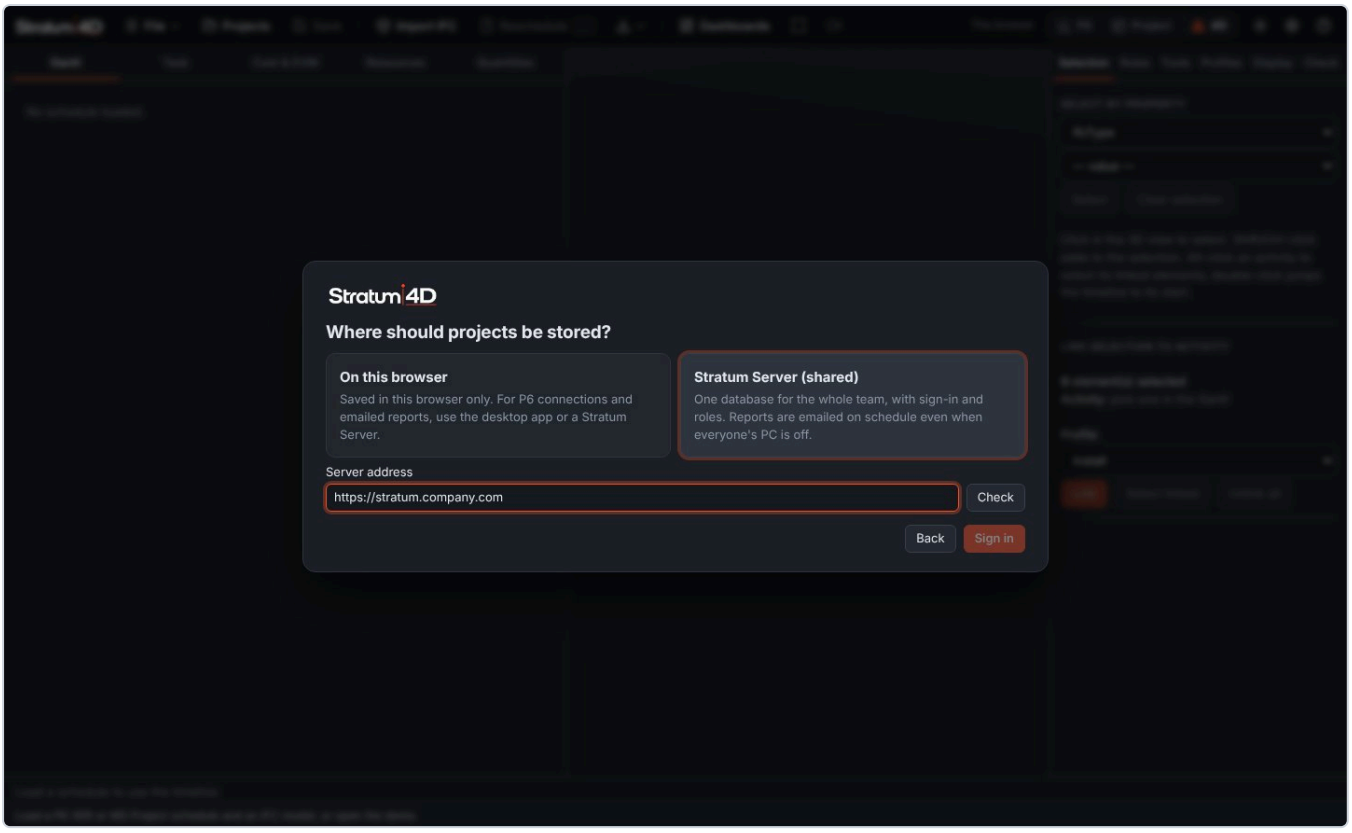


Figure 2. Step 2: where projects are stored. Choosing Stratum Server asks for the server address, then for an account; on a brand-new server the first account becomes the administrator.

Storage. In the browser the choice is between this browser and a Stratum Server. In the desktop app the first option is the local database on your computer. After setup Stratum offers to open the P6 and email settings, since both are usually configured once by an administrator.

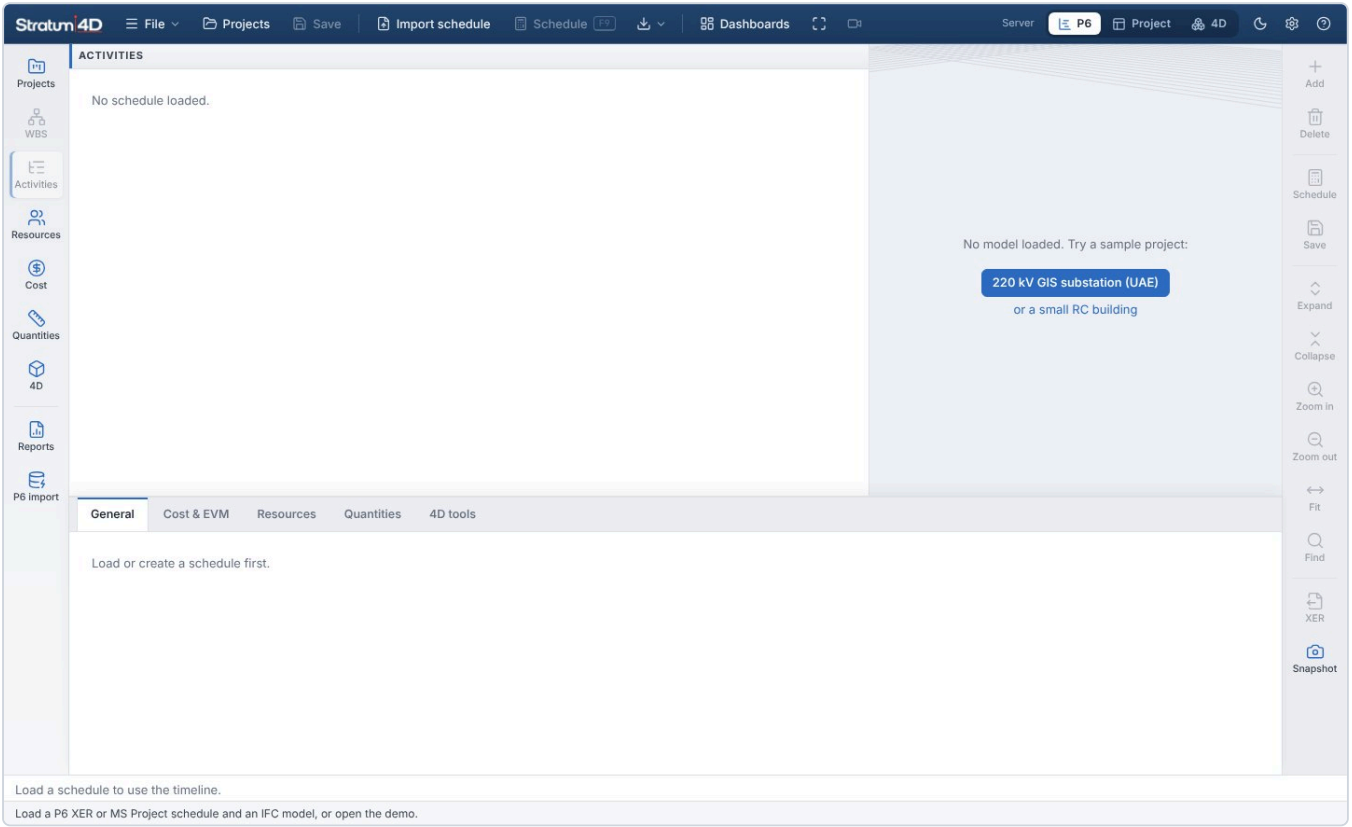


Figure 3. An empty workspace in the P6 layout. The buttons in the middle load one of the two built-in sample projects.

4. The workspace and the three layouts

The top bar holds the commands you use most, the layout switch and the light/dark button. Commands that change wording between layouts (Schedule / Calculate / Reschedule, Activities / Tasks) follow the layout automatically.

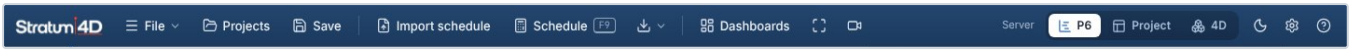


Figure 4. The P6 layout's top bar: File menu, Projects, Save, Import schedule, Schedule (F9), Export, Dashboards, Fit view and Record. On the right: where projects are stored, the layout switch (P6 · Project · 4D), theme, Settings and Help.

Switching layouts

Click **P6**, **Project** or **4D** in the top bar, or press `Ctrl` + `Alt` + `1`, `2` or `3`. The open project, the selection and the playback date stay as they are; only the arrangement, the column set and the vocabulary change.

	P6 style	MS Project style	4D style
Look	Navy title bar, directory bar on the left, command bar on the right	Green title strip and ribbon with large buttons	Graphite dark theme with the Stratum orange
Main area	Activity table and Gantt on top, 3D view beside it, detail tabs below	Timeline strip, Gantt with row numbers and predecessors, task form below	Gantt left, 3D model centre, linking tools right
Gantt colours	Green remaining, blue actual, red critical, yellow baseline	Light-blue task bars, dark progress, red critical	Blue bars, orange-red critical
Wording	Activity ID, Original Duration, Total Float, Data Date	ID, Task Name, Duration, Total Slack, Status Date	Task ID, Planned Duration, Focus date

P6 style

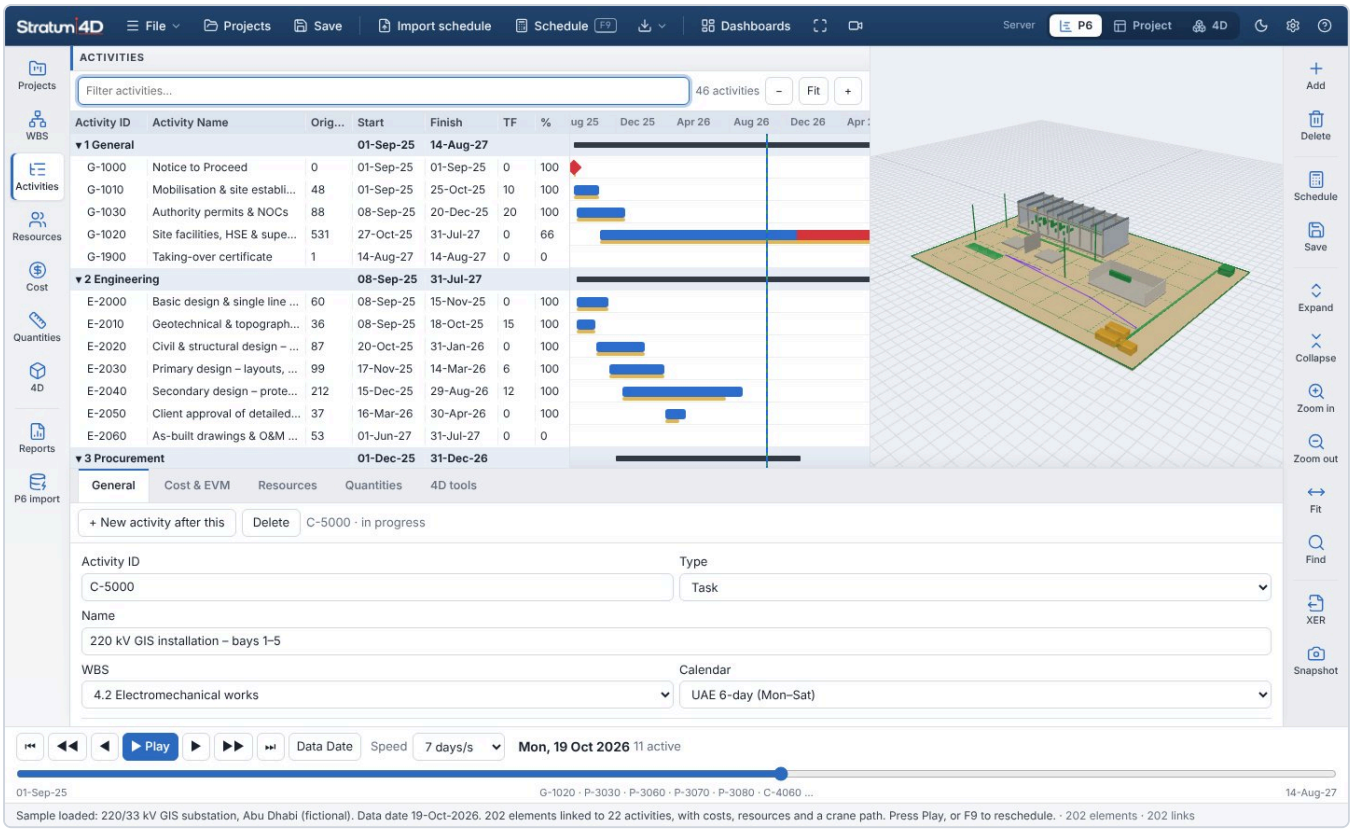


Figure 5. P6 layout with the example project. Activity C-5000 (220 kV GIS installation, bays 1–5) is selected; its details show in the General tab. The vertical line in the Gantt is the data date, 19-Oct-26.

The **directory bar** on the left decides what you look at, top to bottom: Projects, WBS, Activities, Resources, Cost, Quantities, 4D, Reports and P6 import. The **command bar** on the right holds what you do: Add, Delete, Schedule, Save, Expand, Collapse, Zoom in, Zoom out, Fit, Find, XER export and Snapshot.

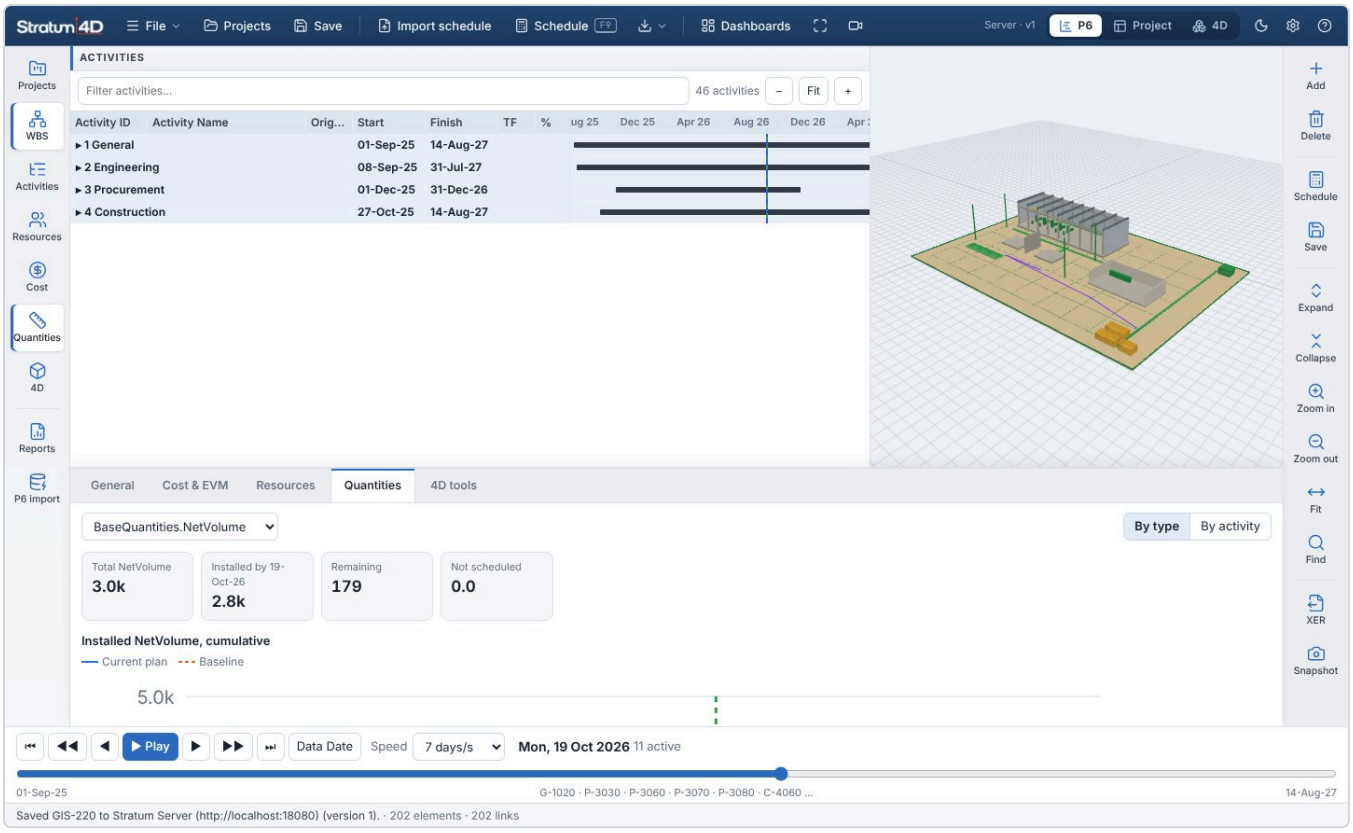


Figure 6. WBS on the directory bar collapses the Gantt to WBS summaries: General, Engineering, Procurement and Construction with their dates. Activities expands it again.

MS Project style

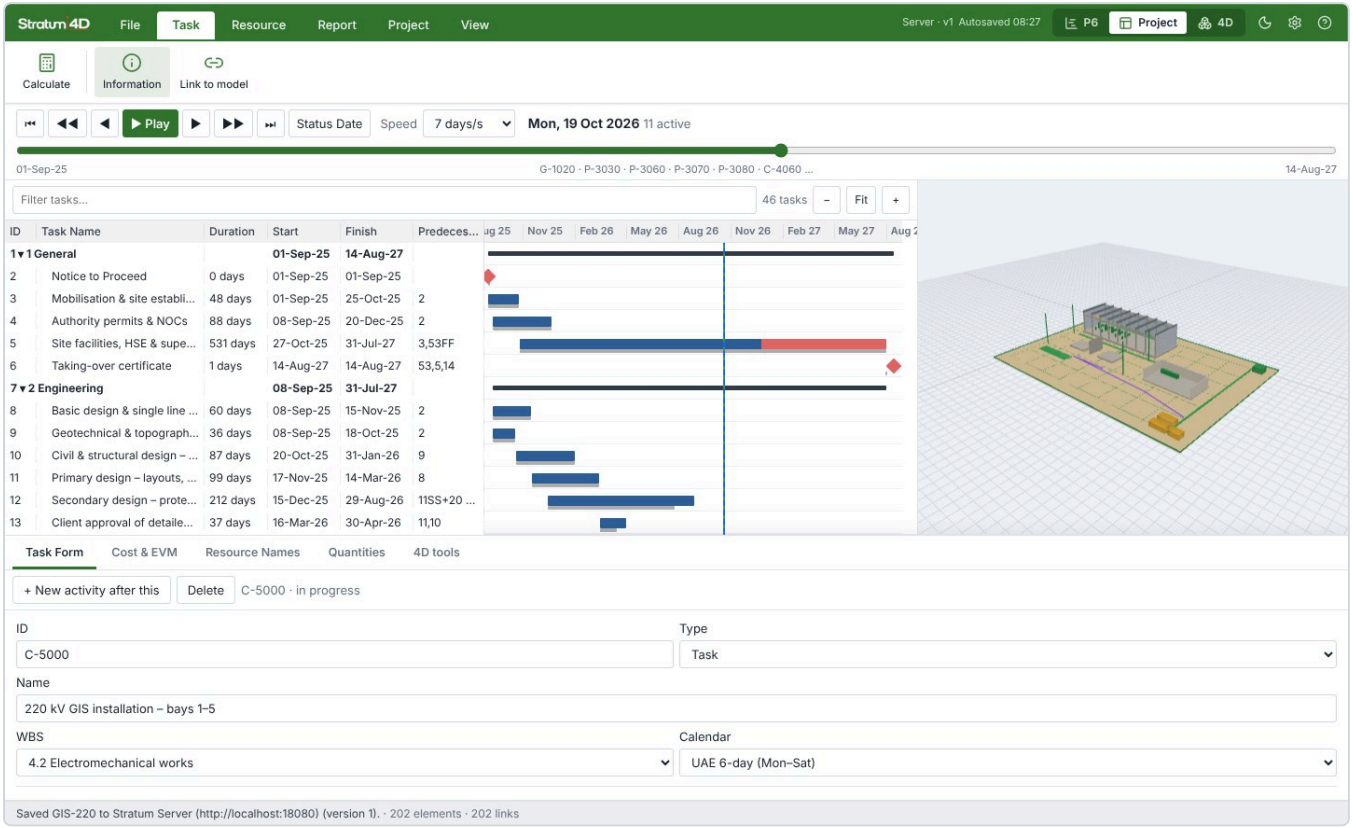


Figure 7. MS Project layout. The ribbon's Task tab is open; the timeline sits above the Gantt, which shows row numbers, durations in days and a Predecessors column.

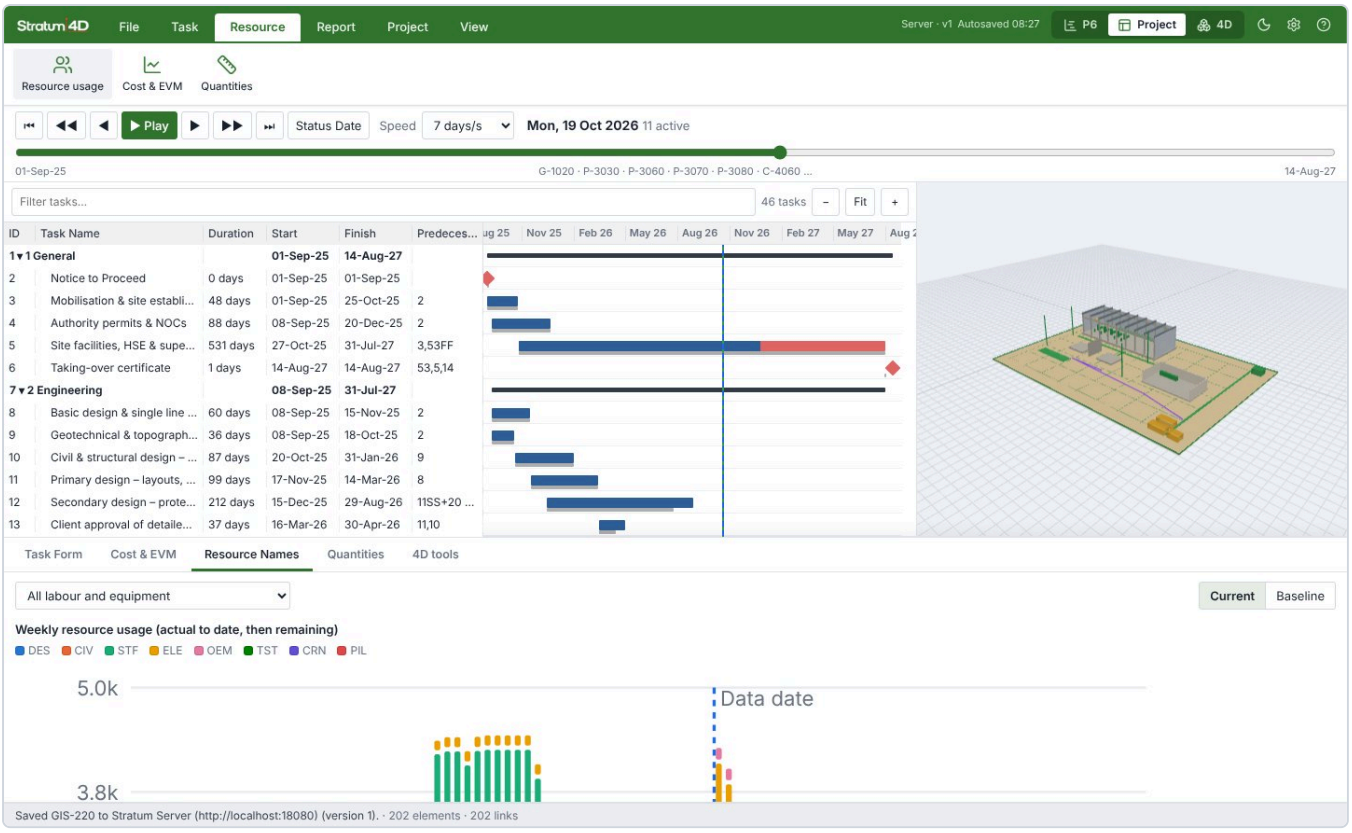


Figure 8. Resource tab of the ribbon: Resource usage opens the weekly histogram in the task form area.

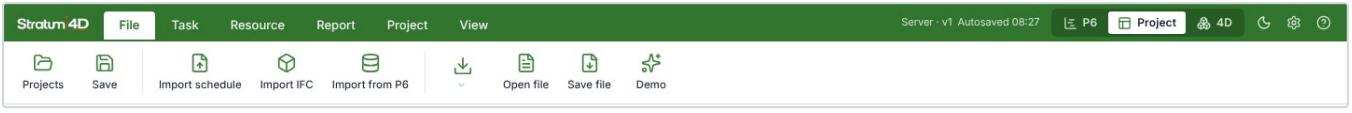


Figure 9. File tab: projects, save, imports, P6 import, export and project files as large ribbon buttons.

4D style

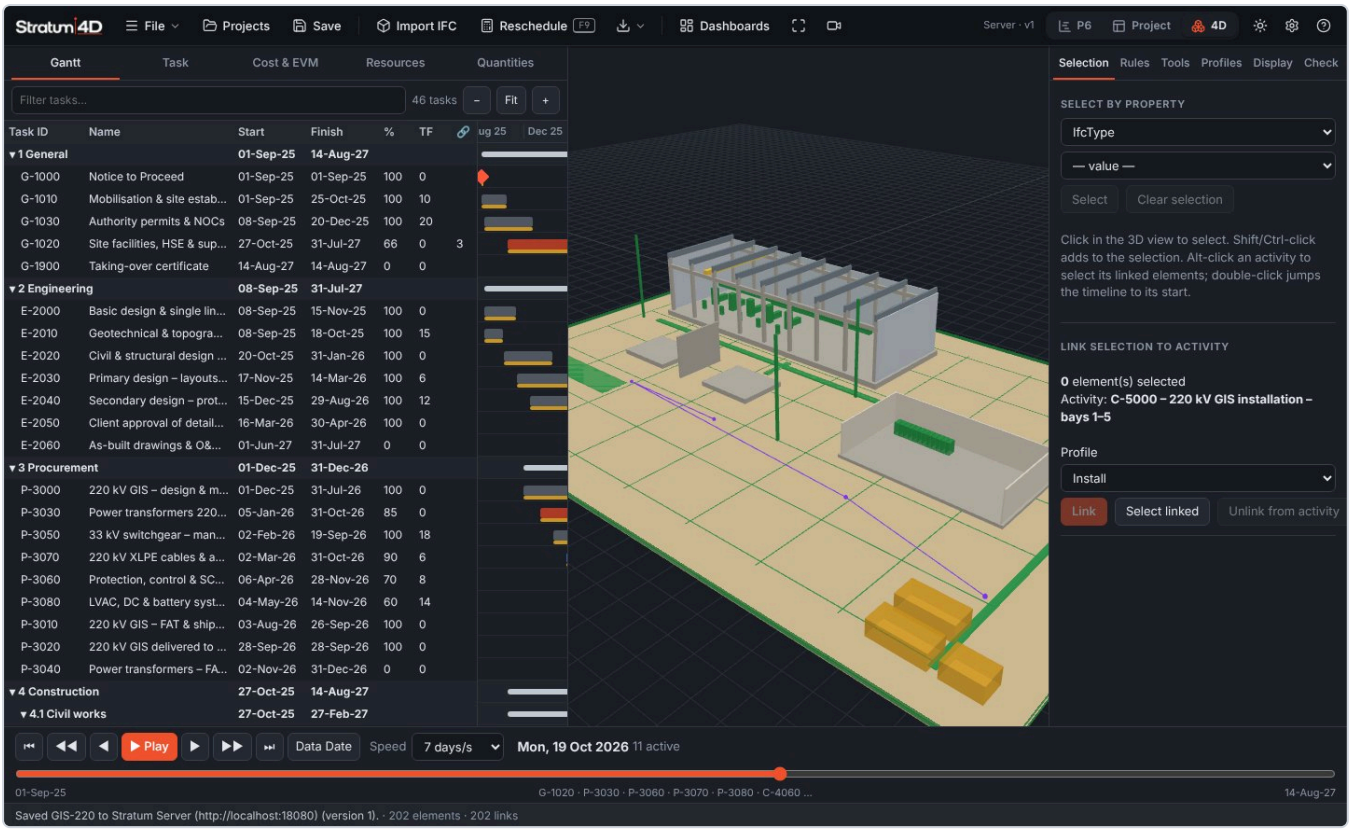


Figure 10. 4D layout in its default dark theme. Green elements are being installed at the data date: the GIS bays, cable trenches and the 33 kV switchgear. The amber boxes are the site cabins, which show only while site facilities (G-1020) run.

Theme, font and text size

The sun/moon button flips between light and dark. *Settings → Interface* has the full set: layout, font, text size, theme (match layout, light or dark) and density.

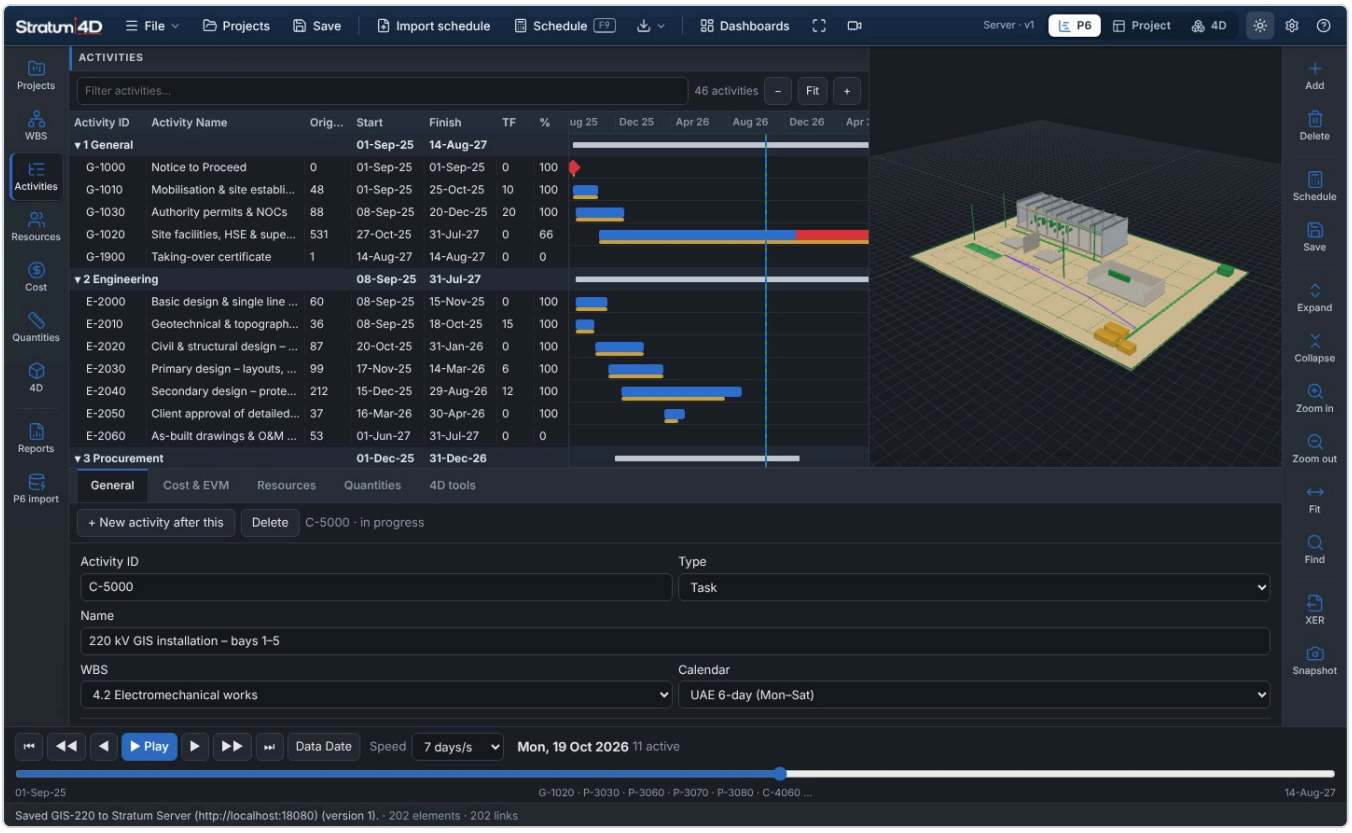


Figure 11. The P6 layout in the dark theme. The 3D view follows the theme.

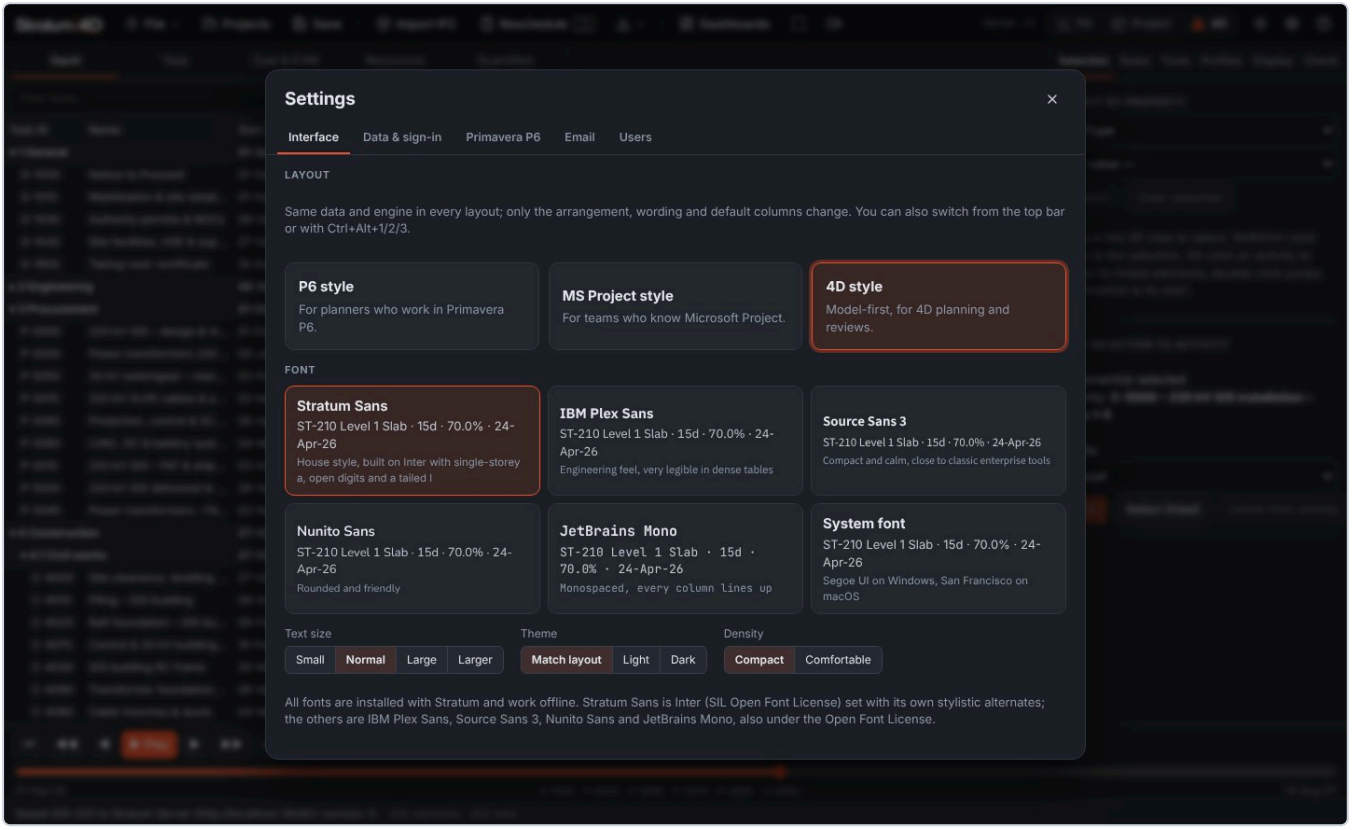


Figure 12. Settings → Interface. Each font shows a sample line in its own face.

Fonts are installed with Stratum, so the desktop app looks the same offline. The default, **Stratum Sans**, is Inter set with its own stylistic alternates (single-storey a, open digits, tailed l). IBM Plex Sans, Source Sans 3, Nunito Sans, JetBrains Mono and the system font are also offered. All are under the SIL Open Font License. Gantt columns widen automatically for larger text or wider fonts.

5. Working with the schedule

Importing

Import schedule reads Primavera P6 XER files and Microsoft Project XML (MSPDI). From XER, Stratum reads WBS, activities, relationships with lags, calendars, constraints, activity codes, user-defined fields, actual and baseline dates, resources, rates and assignments. Re-importing a newer XER over the open project updates it: links to the model are kept by Activity ID, and the Check tab lists what changed.

The Gantt

Bars are drawn from the current dates with the baseline as a thin yellow bar underneath. The dashed vertical line is the data date and the solid line is the playback date. Click a WBS row to fold it; use the filter box to find activities by ID or name; `-`, *Fit* and `+` zoom the timescale. Alt-click an activity to select its linked model elements; double-click to move the timeline to its start.

Editing activities

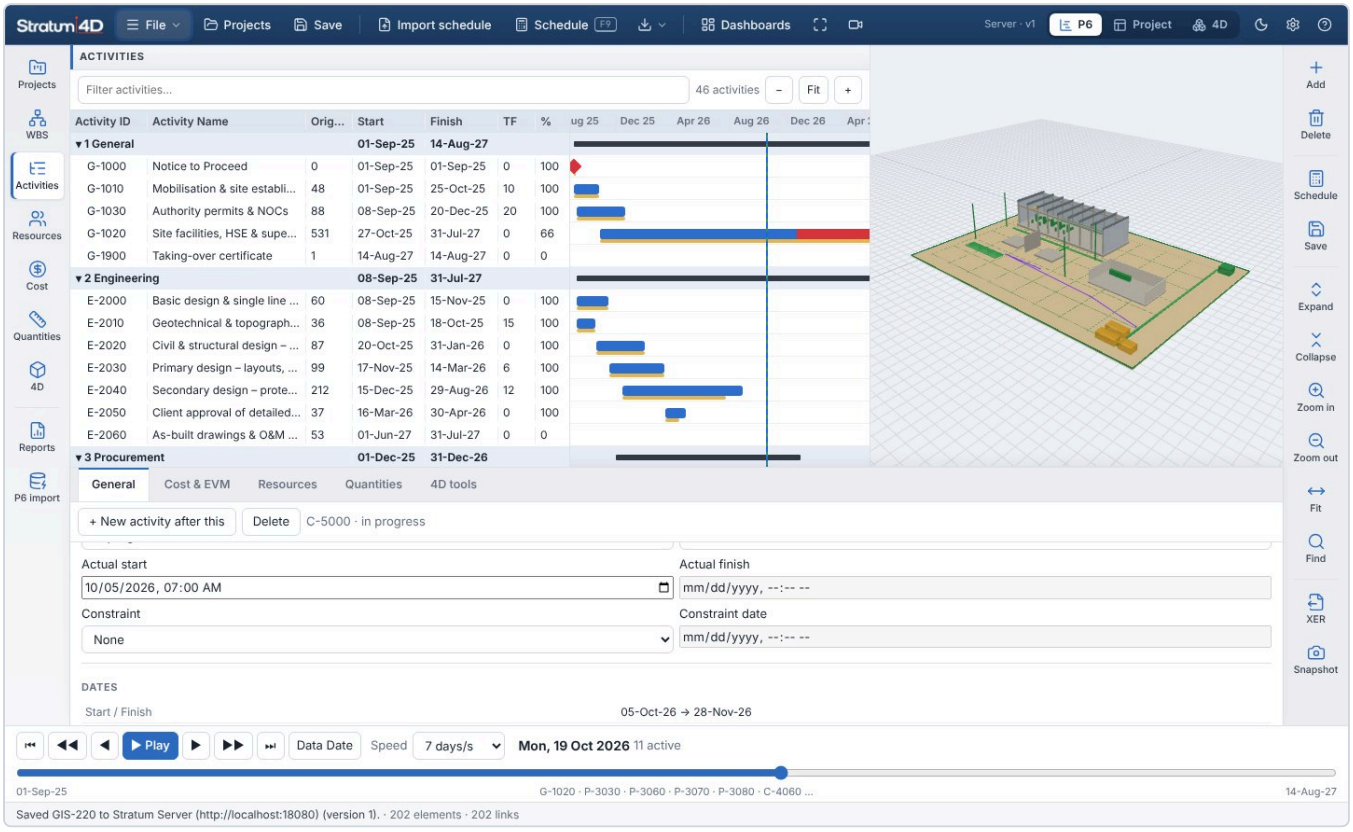


Figure 13. The General tab for C-5000: ID, type, name, WBS and the UAE 6-day calendar, with durations, status, actual dates, constraints, user fields, relationships and resource assignments further down.

Add an activity after the selected one with *Add* (or *+ New activity after this*). It gets a 5-day duration and a finish-to-start link from the selected activity. *Delete* removes it with its relationships. Edits mark the schedule as changed and a banner asks you to reschedule.

Scheduling (F9)

Press `F9` or *Schedule*. The CPM engine follows P6's default options:

- FS, SS, FF and SF relationships with lags, on each activity's working calendar and holidays;
- nine constraint types;
- retained logic from the data date;
- total and free float, critical path, level-of-effort and WBS summary dates;
- loop detection and open-end warnings.

For the example project F9 finishes on 12-Aug-27, with the critical path running through transformer manufacture, delivery, erection and energisation.

Exporting

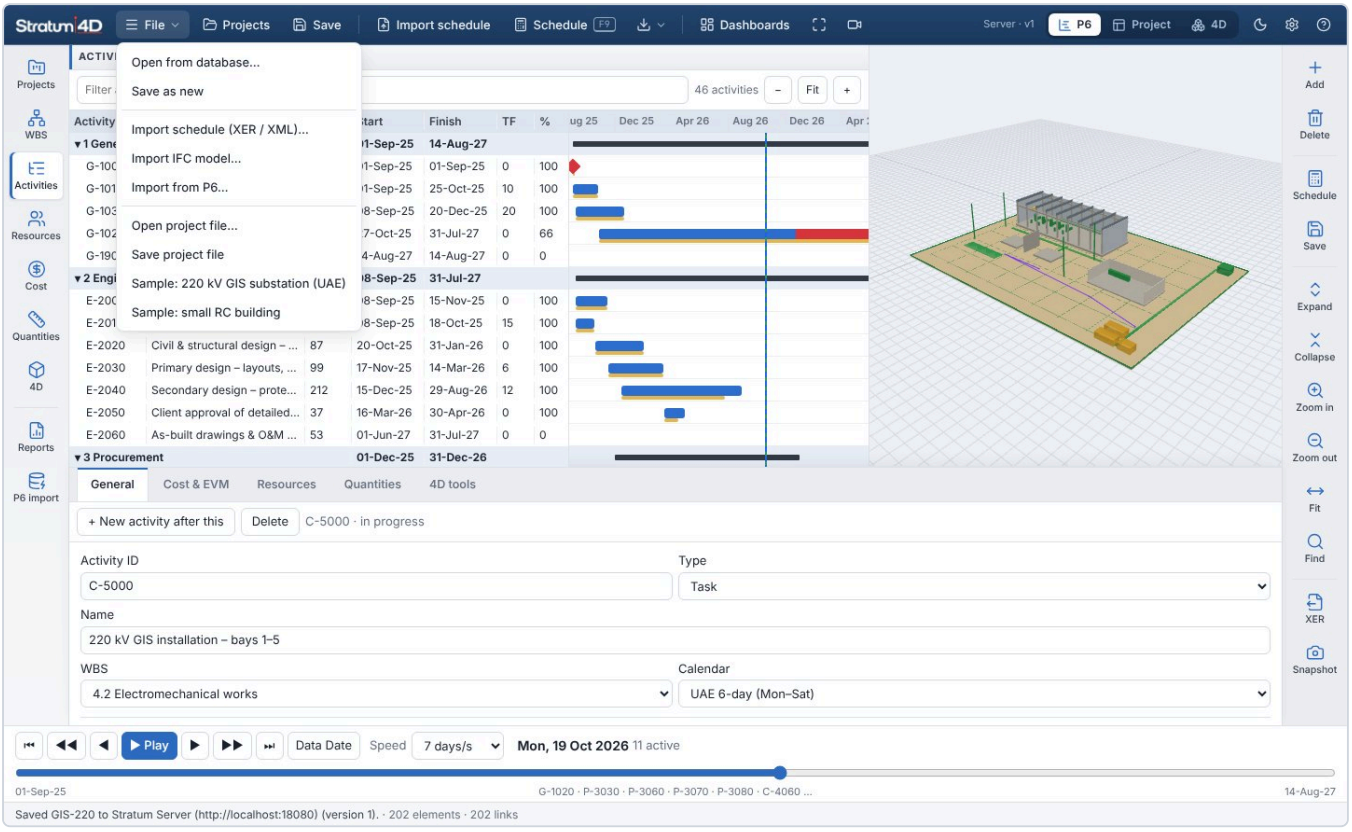


Figure 14. The File menu: database, imports (schedule, IFC, P6), project files and the samples.

Export → P6 XER writes a file that updates the original project when imported in P6 with *Update Existing Project*; tables Stratum did not touch pass straight through. *CSV* opens in Excel, and *Snapshot PNG* saves the 3D view.

6. 4D simulation

A 4D model links each model element to the activity that builds, removes or uses it, then plays the schedule against the model.

Linking the example model

The example model carries the activity ID in each element's property `Pset_Stratum.ActivityID` (for instance, every part of GIS bays 1–5 holds `C-5000`). One rule therefore links all 202 elements: *element property Pset_Stratum.ActivityID equals Activity ID*.

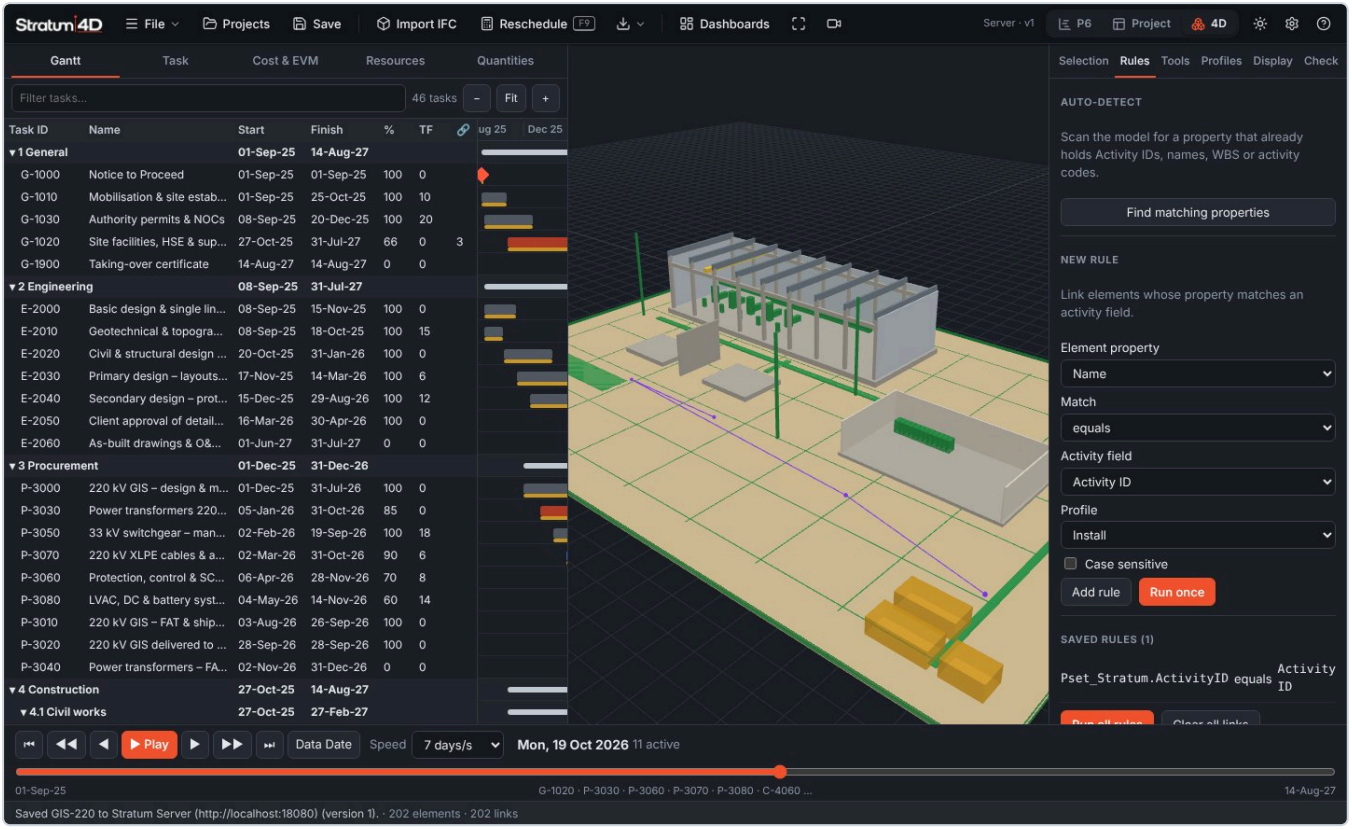


Figure 15. Rules tab. Find matching properties scans the model for a property that already holds activity IDs, names, WBS or codes. Below it, a rule is built from an element property, a match type, an activity field and a profile.

For your own models, the same approach works with a shared parameter from Revit, ArchiCAD or Tekla. Without one, select elements in 3D (or *Selection → Select by property*), pick the activity in the Gantt, choose a profile and press *Link*.

Appearance profiles

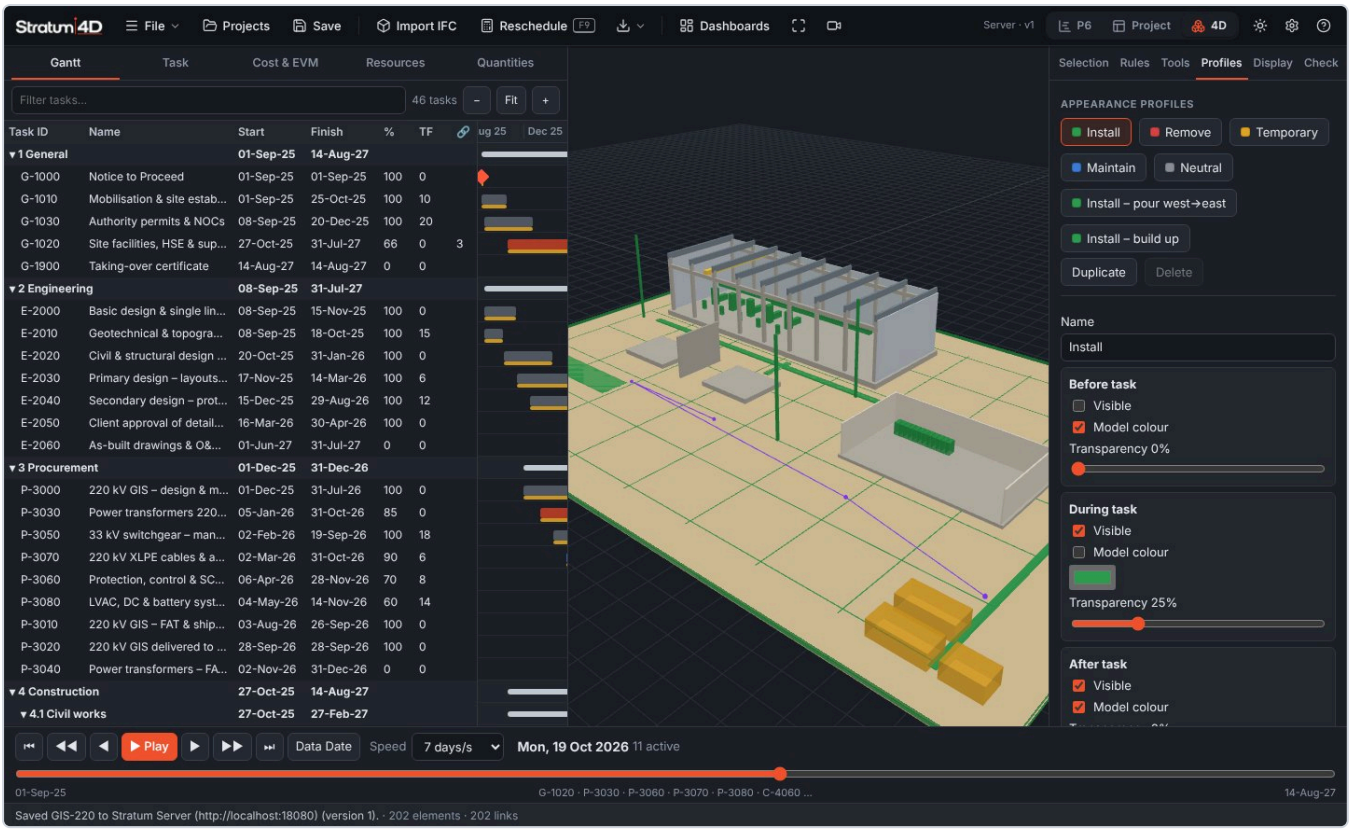


Figure 16. Profiles tab. Each profile sets visibility, colour and transparency before, during and after its activity.

The example uses *Install* for equipment, *Install – pour west→east* for rafts, slabs and trenches, *Install – build up* for walls, columns, piles and cladding, and *Temporary* for the site cabins and the 200 t mobile crane, which appear only while their activity runs.

Groups, pours and equipment paths

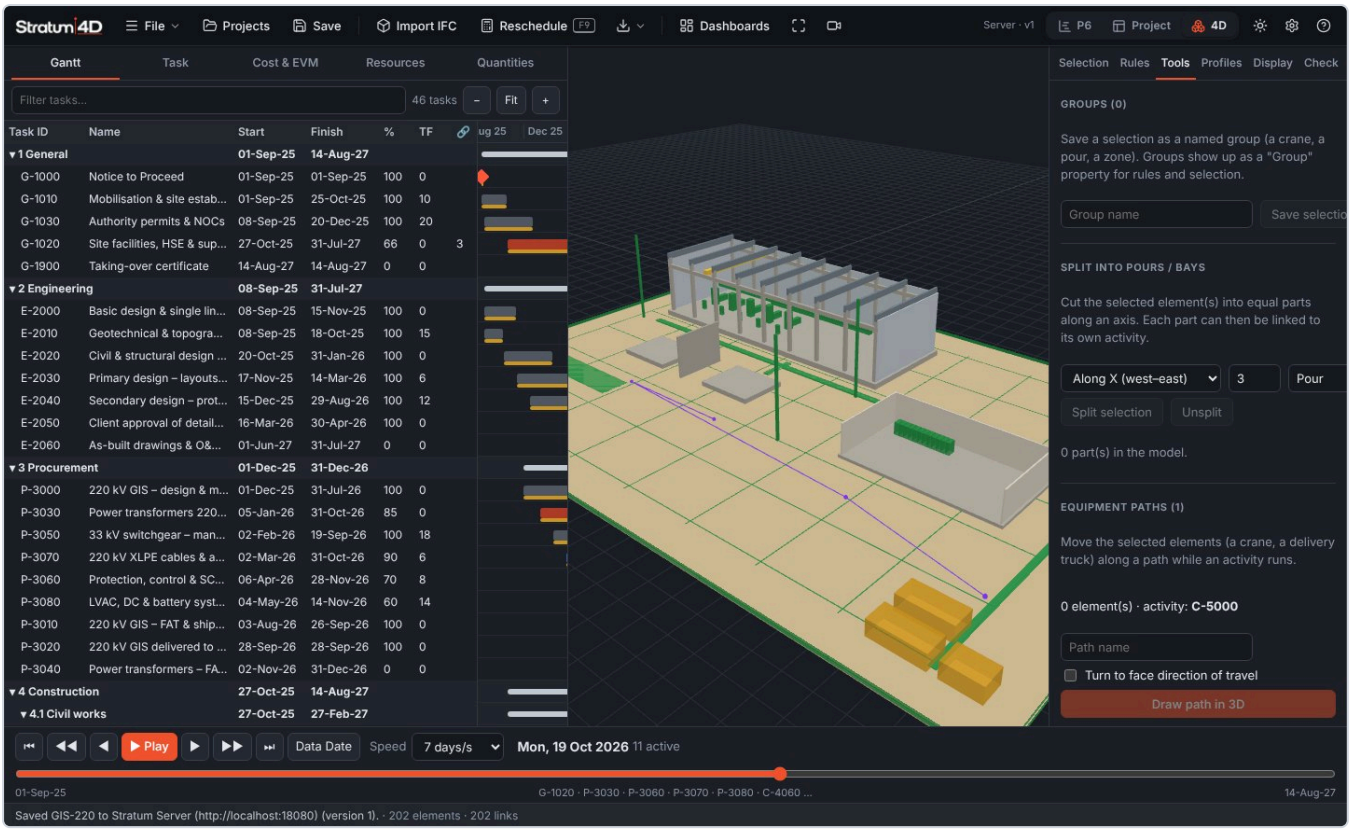


Figure 17. Tools tab: save selections as named groups, split elements into pours or bays, and move equipment along a path.

The example's crane path takes the 200 t mobile crane from the gate along the access road to transformer T1 and then T2 during activity C-4900 (transformer placing).

Display options and checks

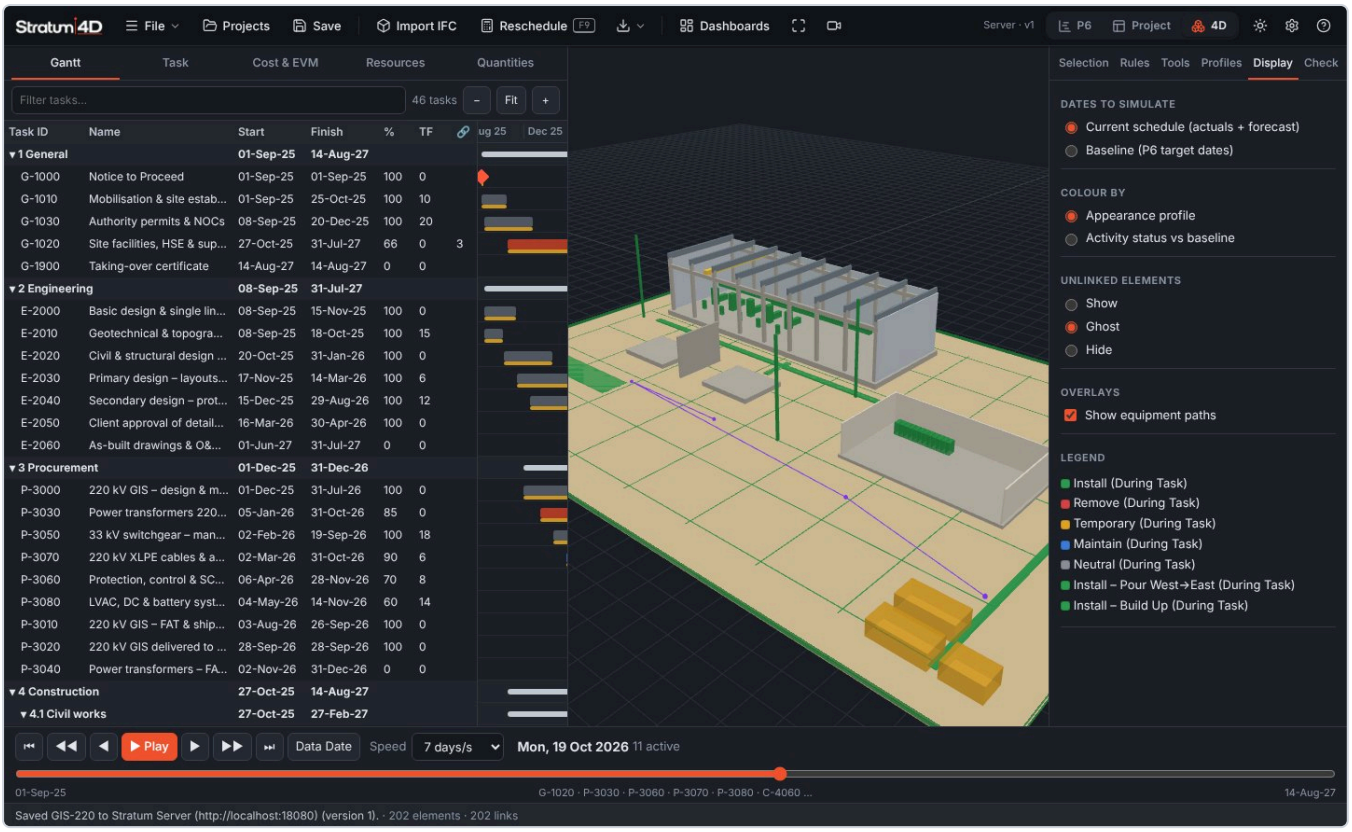


Figure 18. Display tab: simulate current dates or the P6 baseline, colour by profile or by status against baseline, ghost or hide unlinked elements, and show equipment paths.

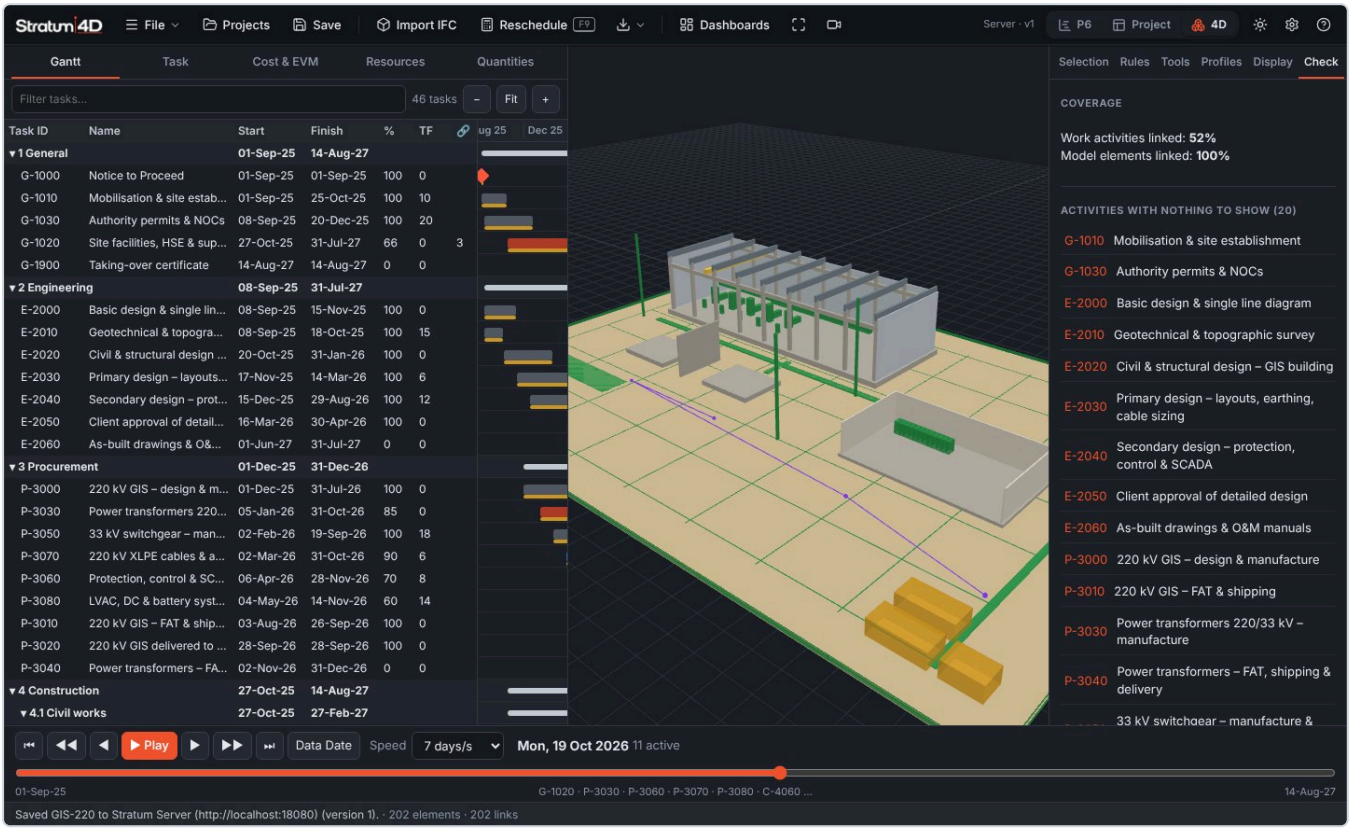


Figure 19. Check tab: link coverage, activities with nothing to show, elements never shown, and the change report after a schedule update.

Playing the sequence

Use the timeline at the bottom: Play/Pause (`Space`), step one day (`←` `→`) or one week (`Shift`+arrows), jump to the start, data date or finish, and set the speed. *Record* saves the playback as a WebM video.

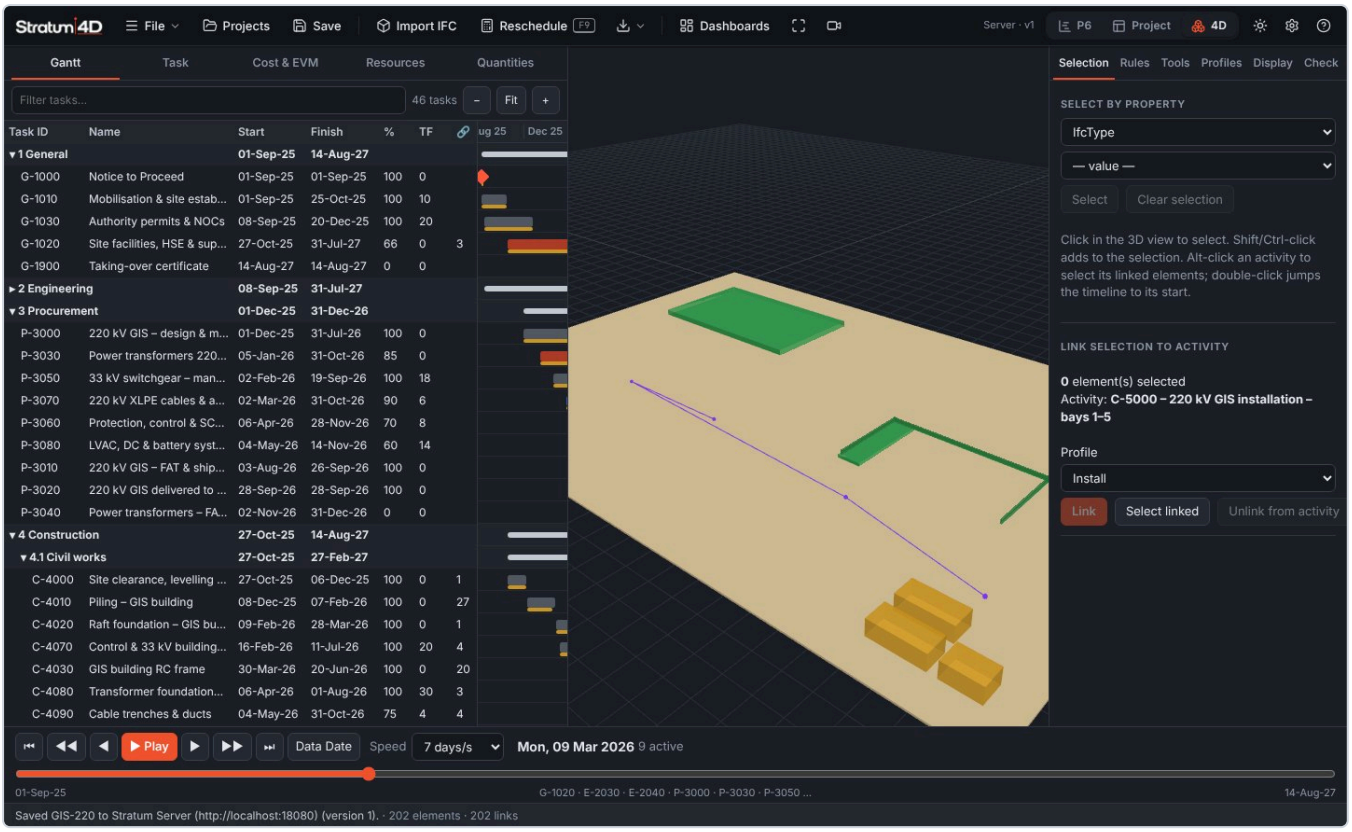


Figure 20. 9 March 2026: site levelled, piles done, the GIS building raft being poured (green) and the control building foundations started. The amber boxes are the site cabins.

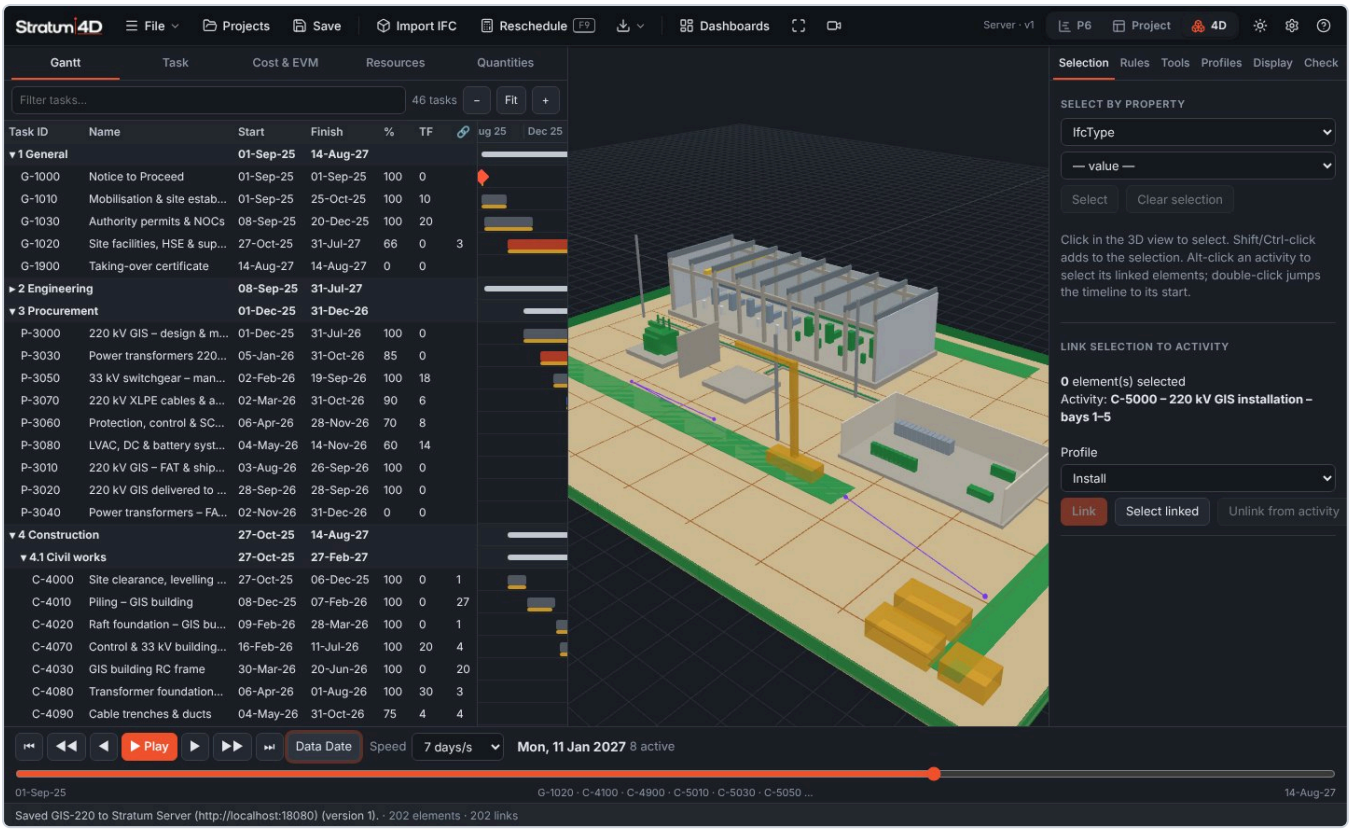


Figure 21. 11 January 2027: GIS hall complete with bays being installed, transformer T1 being erected, the crane (yellow) on its path, cables going into the trenches and the boundary wall under way.

7. Cost, resources and quantities

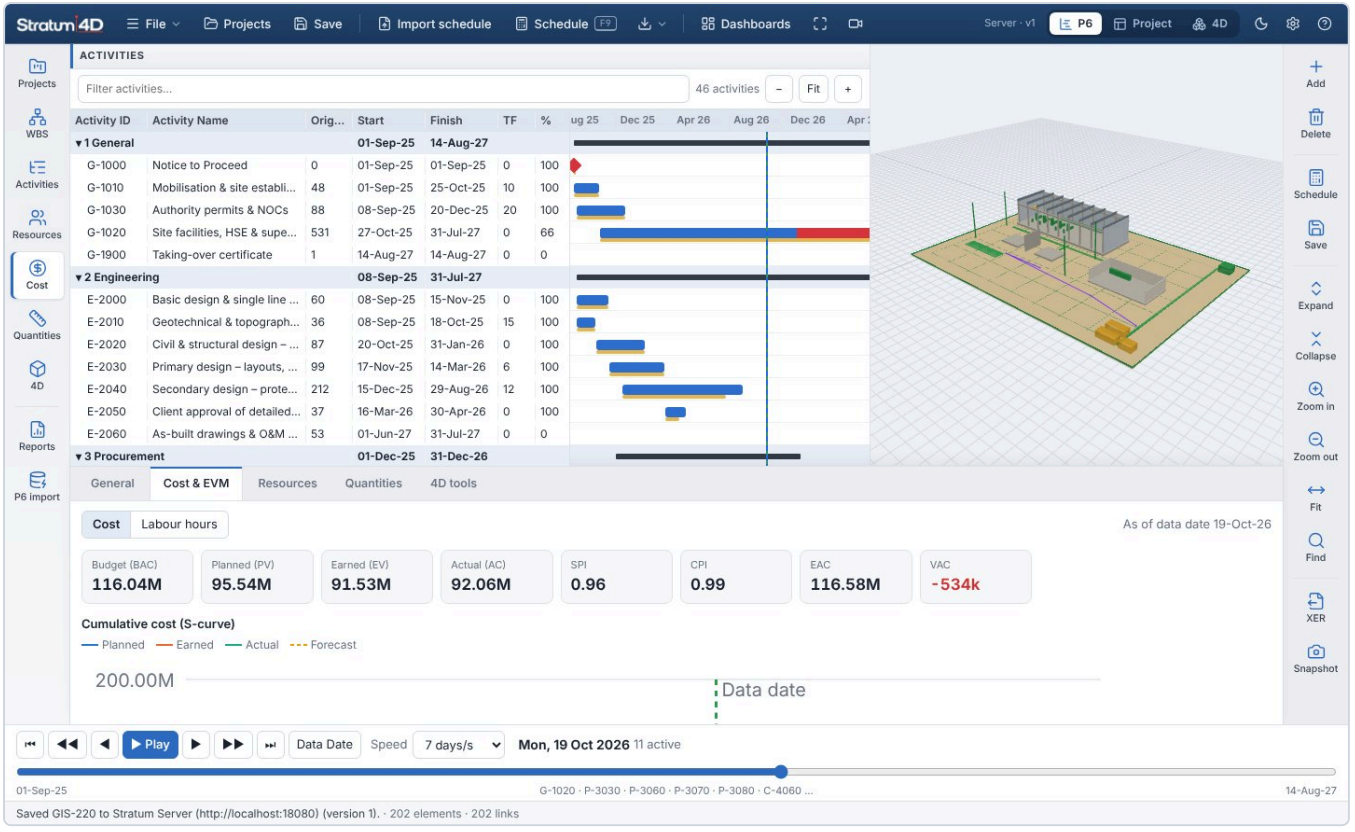


Figure 22. Cost & EVM tab for the example at the data date: budget at completion AED 116.04M, planned value 95.54M, earned value 91.53M, actual cost 92.06M, SPI 0.96, CPI 0.99, EAC 116.58M and VAC −534k.

Earned value is calculated from resource assignments in cost or labour hours. The S-curve shows planned, earned, actual and forecast; a WBS table below rolls the figures up. In the example, SPI 0.96 reflects the late GIS delivery and transformer manufacture, and the small cost overrun comes from piling, cladding and secondary design.

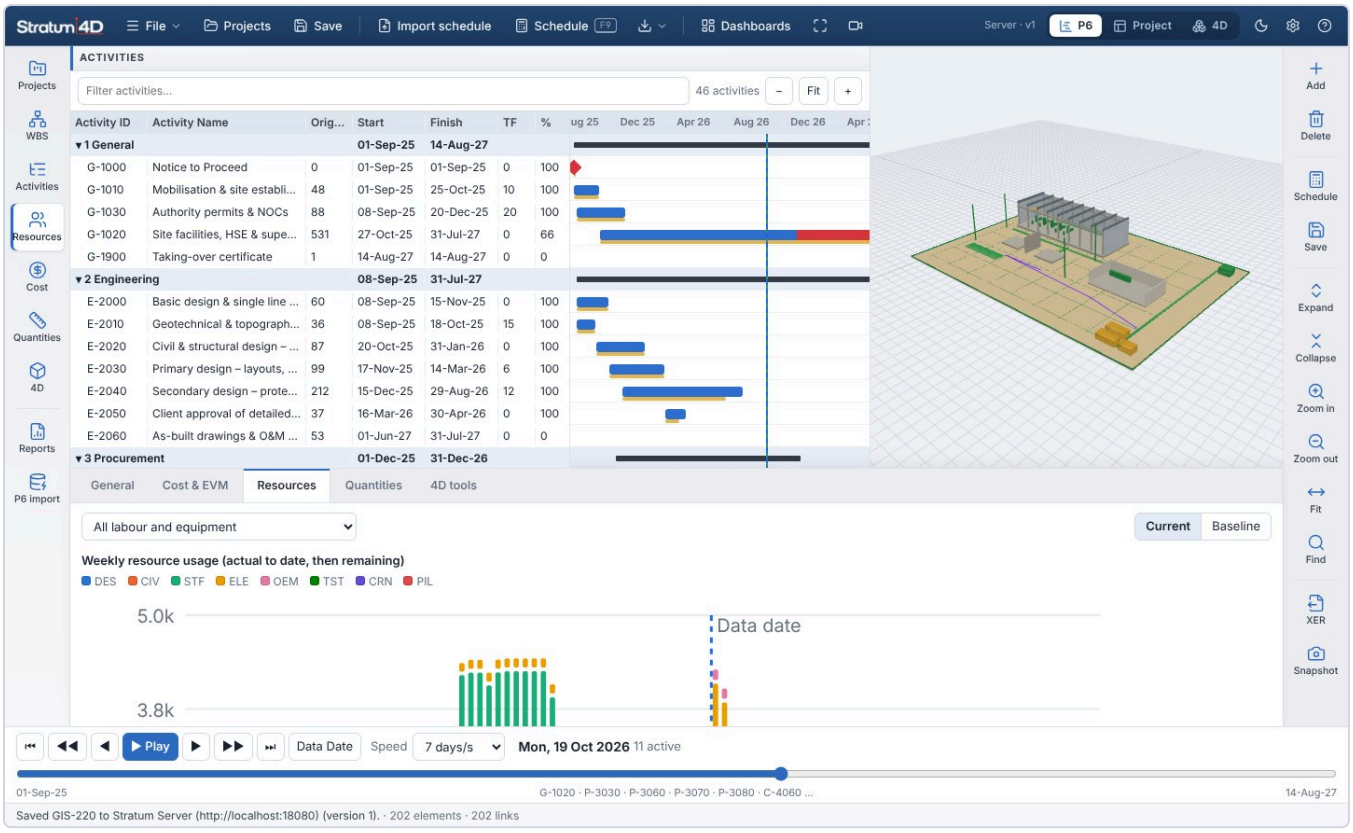


Figure 23. Resources tab: weekly usage per resource, actual to date then remaining (or the baseline), filtered to labour and equipment or a single resource.

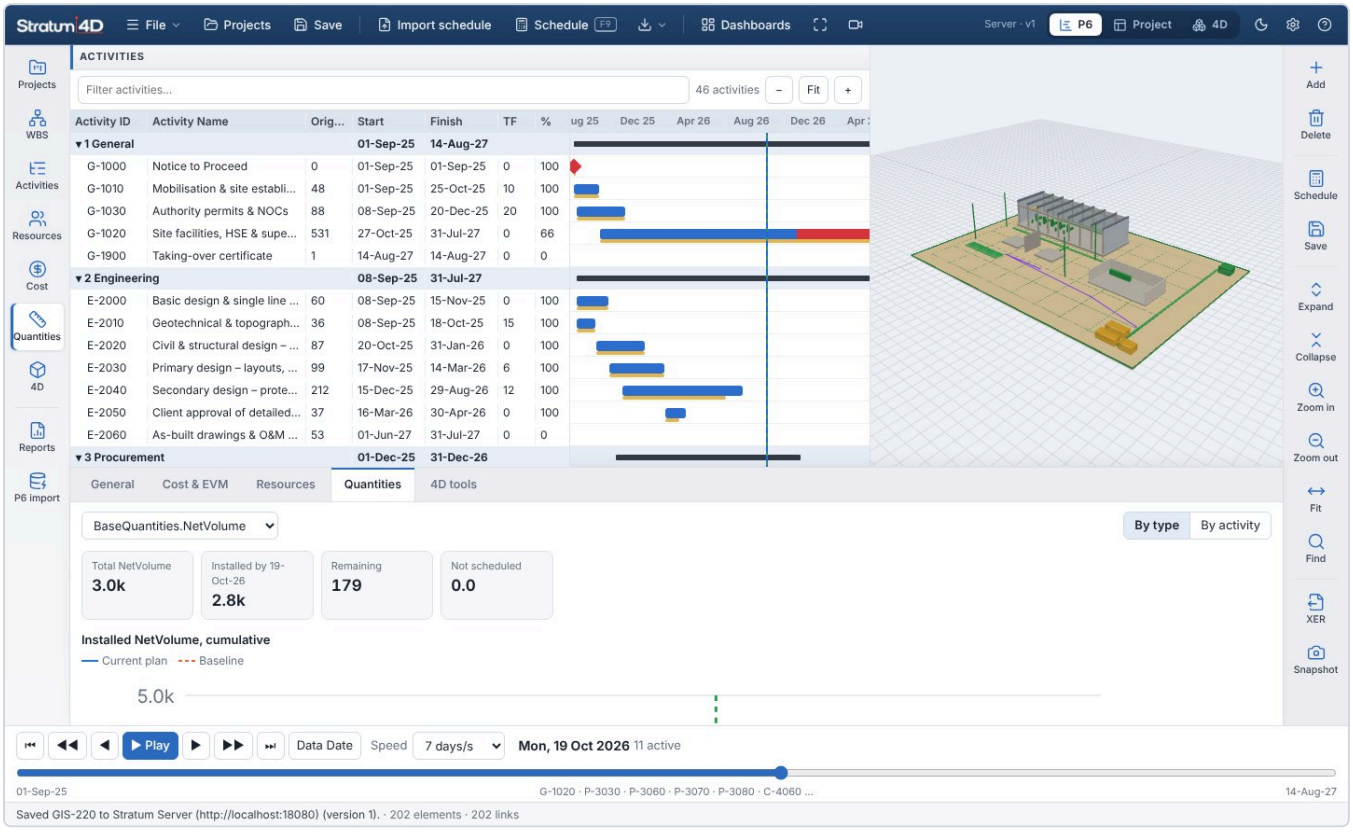


Figure 24. Quantities tab: any numeric IFC quantity (here NetVolume) installed by the playback date, as tiles and a cumulative curve against the baseline, grouped by IFC type or activity.

8. The project database

Save (`Ctrl`+`S`) stores the open project in the database: schedule, model links, rules, profiles, groups, pours and paths. Every save gets a version number, shown in the top bar. If someone else saved the project after you opened it, Stratum refuses to overwrite their work and offers *Save as new* instead.

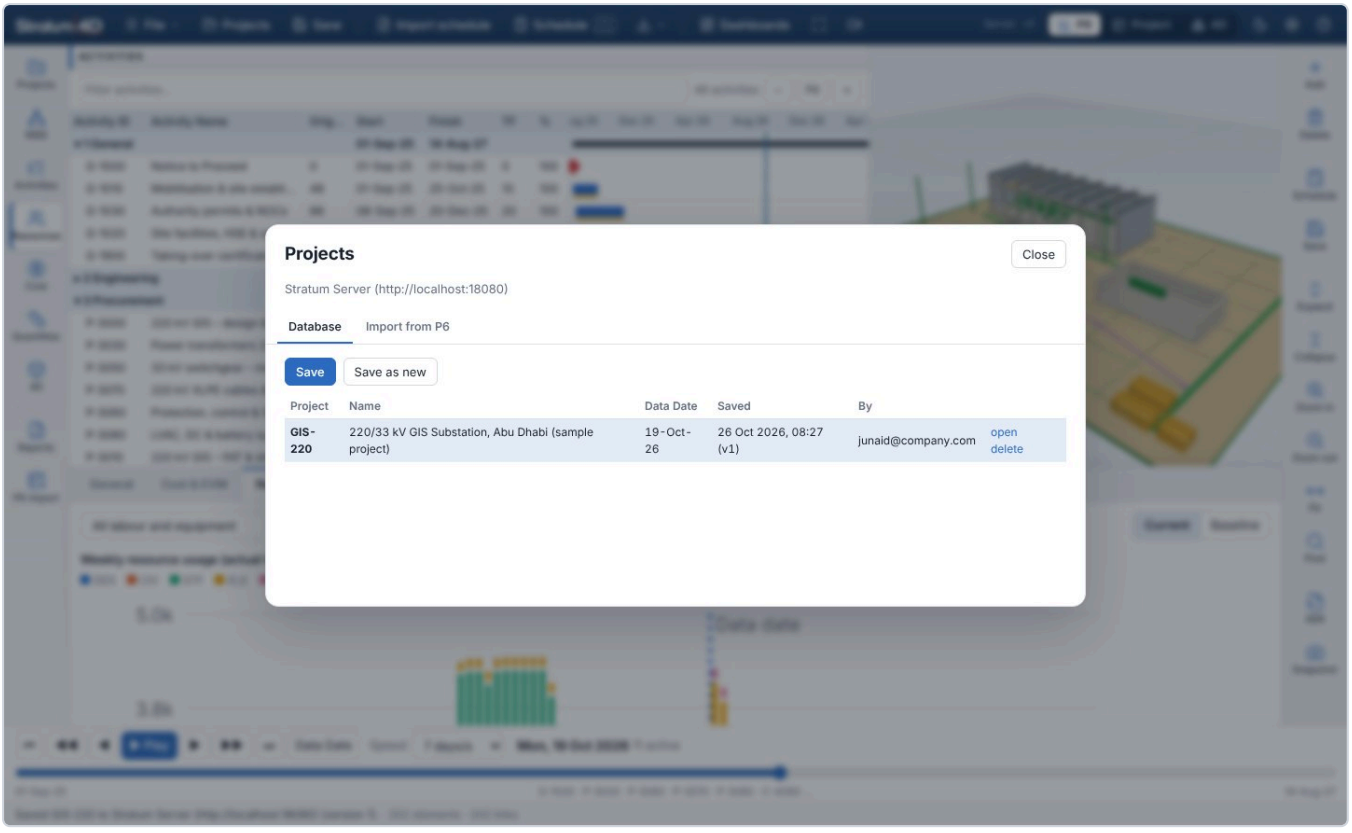


Figure 25. Projects → Database: open, save, save as new or delete. The current project is highlighted with its version.

The model geometry is not stored in the database; open the IFC again after opening a project and the links match by IFC GlobalId. `.stratum.json` project files are still available from the File menu for sharing outside the database.

9. Primavera P6 connections

The desktop app and Stratum Server can talk to P6 directly. The browser cannot, so use XER files there. Connections are set up once in *Settings → Primavera P6*, usually by an administrator.

Connection type	Reads	Writes back	Notes
P6 EPPM (REST API)	Projects, WBS, activities, relationships, calendars, codes, UDFs, resources, assignments	Actual start/finish, % complete, remaining duration	Address normally ends in /p6ws/restapi. Username/password or a bearer token (Oracle-hosted P6).
P6 Professional database (SQL Server, Oracle, SQLite)	The same tables as an XER export	No (read-only by design)	Use a read-only database login; send changes back by XER.

Checking a connection

Test connection checks each link in the chain and stops at the first one that breaks, with a plain-language fix. *Copy report for IT* puts the result on the clipboard.

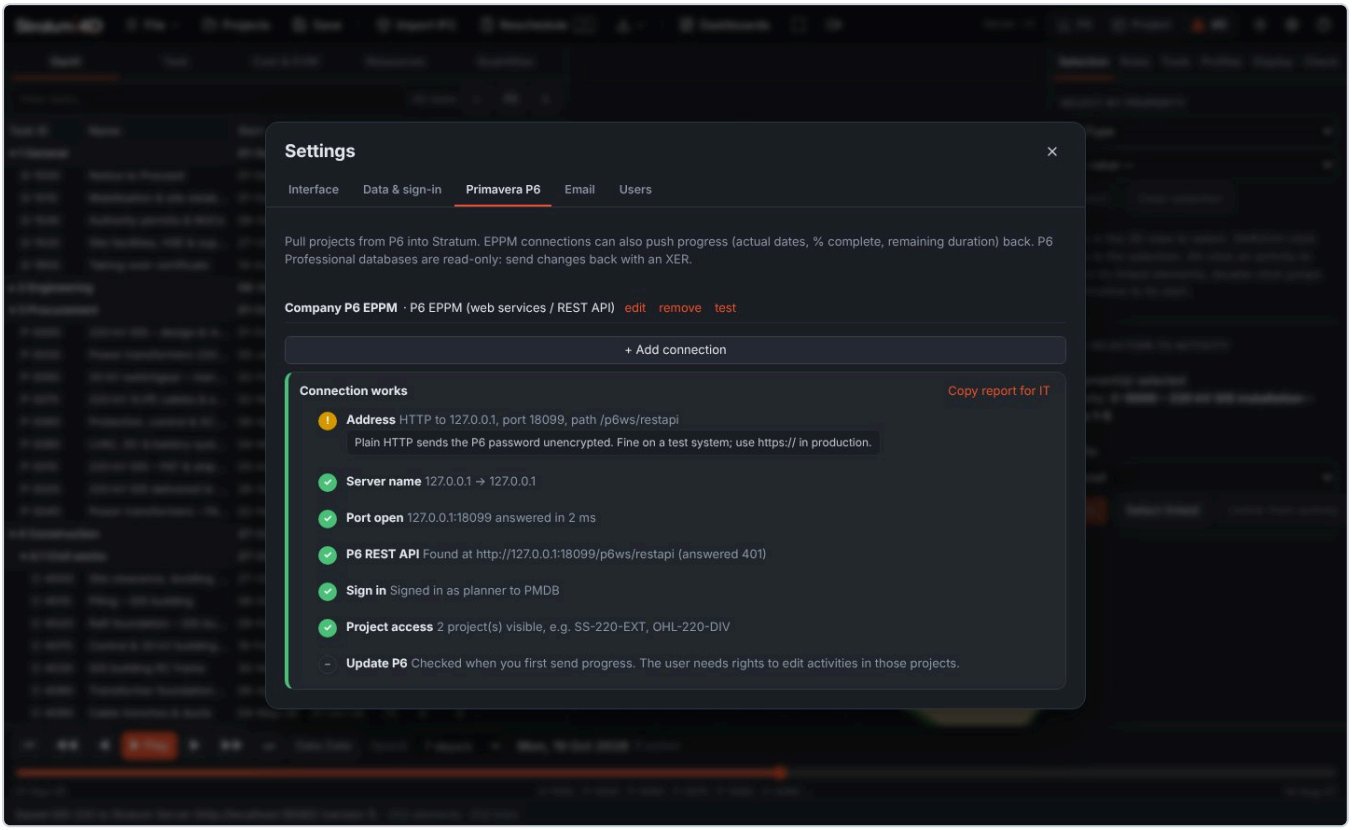


Figure 26. A successful check: address, server name, port, P6 REST API, sign-in and project access.

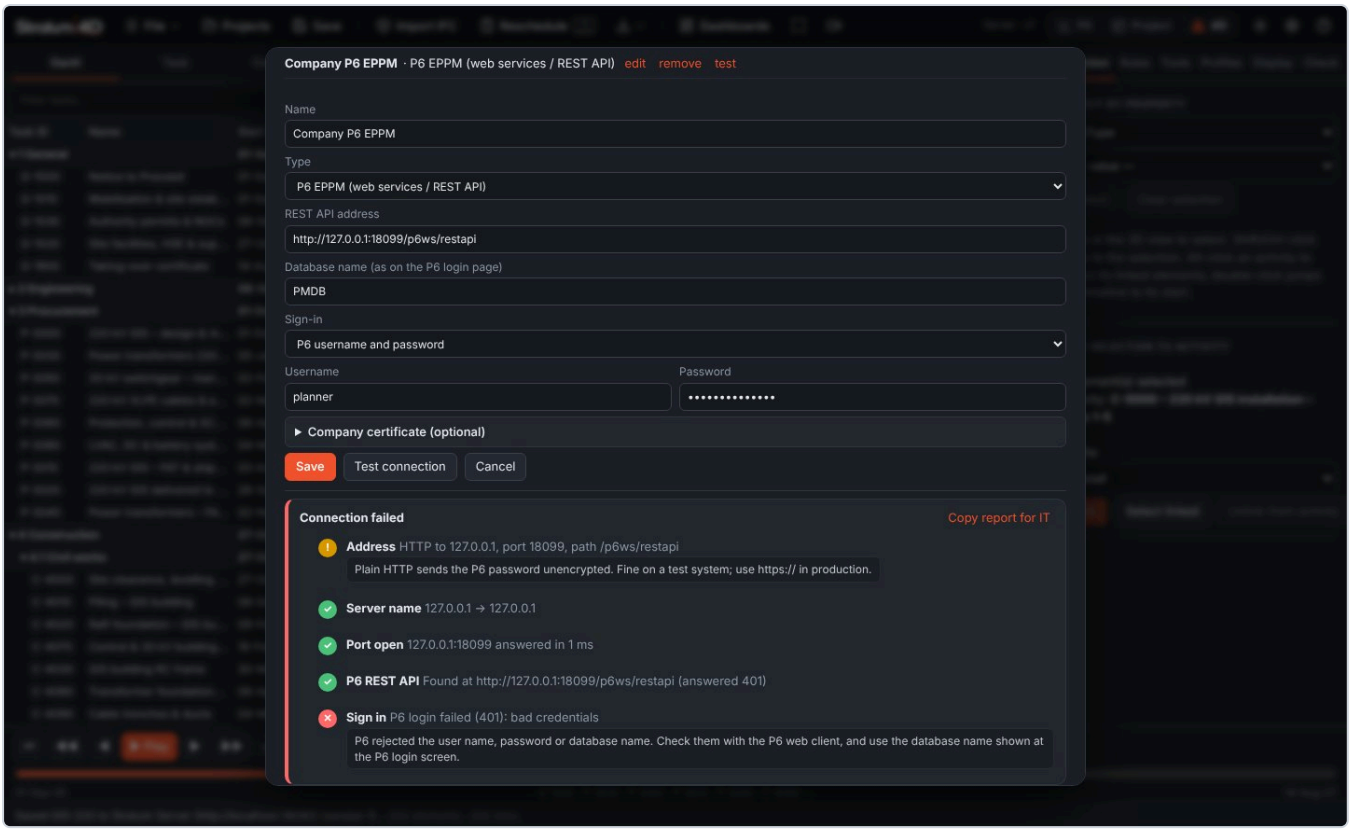


Figure 27. A failed check: sign-in rejected, with what to check.

Step	What it catches
Address	Not a web address; plain http:// (password sent unencrypted); path not ending in /restapi
Server name	Name not found: spelling, or not on the company network / VPN
Port open	Connection refused (P6 stopped, wrong port) or no answer (firewall, VPN)
Certificate	Expired, issued for a different name, or from a company authority this machine does not trust
P6 REST API	Nothing at this path (REST not deployed), or an application-server error
Sign in	Wrong user, password or database name; user without API access; single sign-on
Project access	Signed in but no projects visible: give the user OBS access

Network and certificates

Stratum only makes outgoing HTTPS calls to P6, so nothing needs port forwarding on the Stratum side. The machine running Stratum must reach the P6 server the same way a browser reaches the P6 web client. If Stratum Server runs in the cloud and P6 on company servers, connect them with a VPN or private link rather than exposing P6 to the internet.

- **Desktop app:** P6 calls use the operating system's network stack, so certificates installed by IT in Windows or the macOS keychain are trusted, and the system proxy is used.
- **Stratum Server:** set `NODE_EXTRA_CA_CERTS` to a file holding the company root certificate, and `HTTPS_PROXY` / `NO_PROXY` if a proxy is needed.
- **Either:** paste the root certificate into *Company certificate (optional)* on the connection.

Importing and updating P6

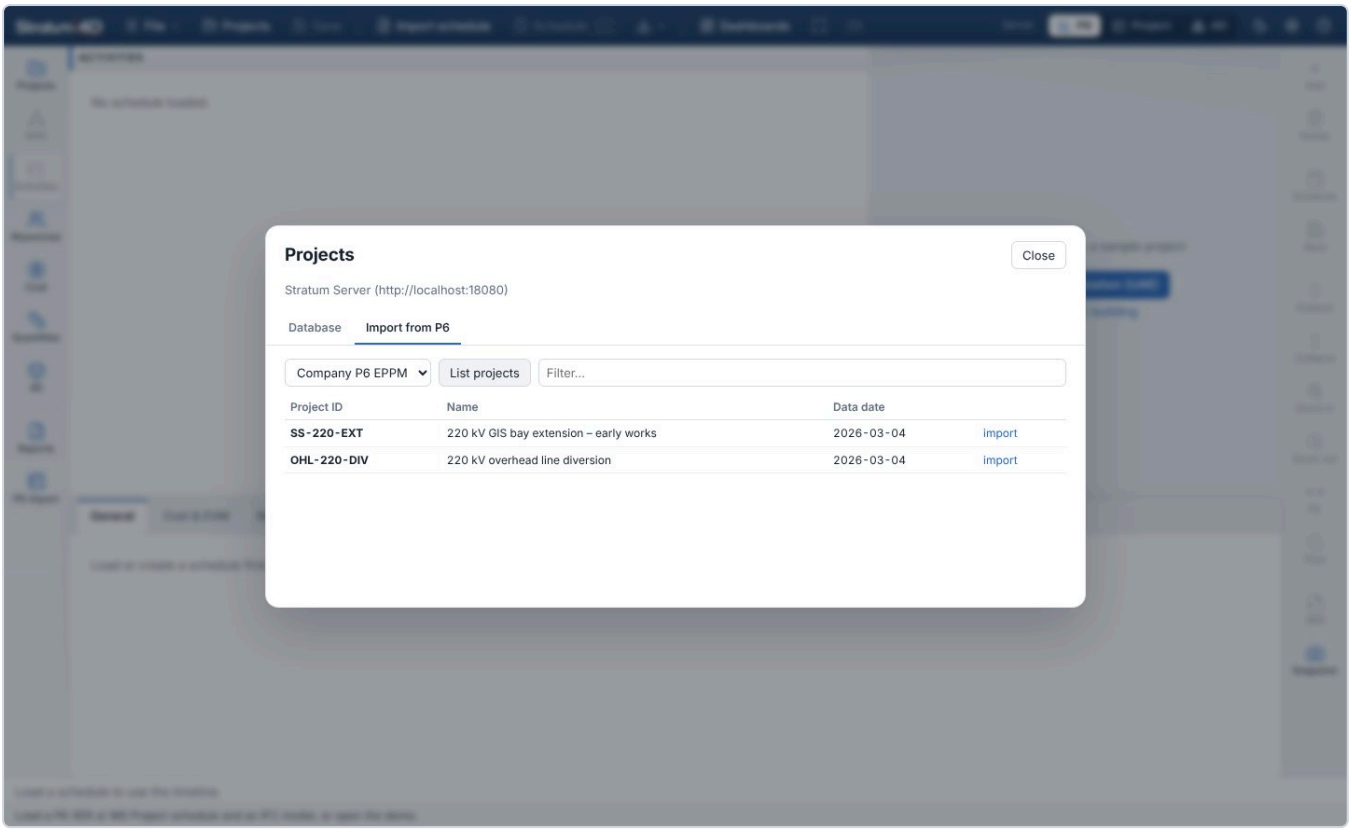


Figure 28. Projects → Import from P6: list the projects on a connection and import one.

An imported project behaves like any other: link it to the model, record progress and report on it. For EPPM projects the top bar shows *Update P6*, which sends actual dates, % complete and remaining duration back for activities that exist in P6. Activities added in Stratum are listed, not pushed. After the update, reschedule in P6 with F9.

10. Dashboards and reports

Dashboards opens a report builder in the spirit of Power BI. Start from a template, then add, resize, drag and configure widgets. Each dashboard shows the open project or a saved project chosen at the top.

The EPC progress report

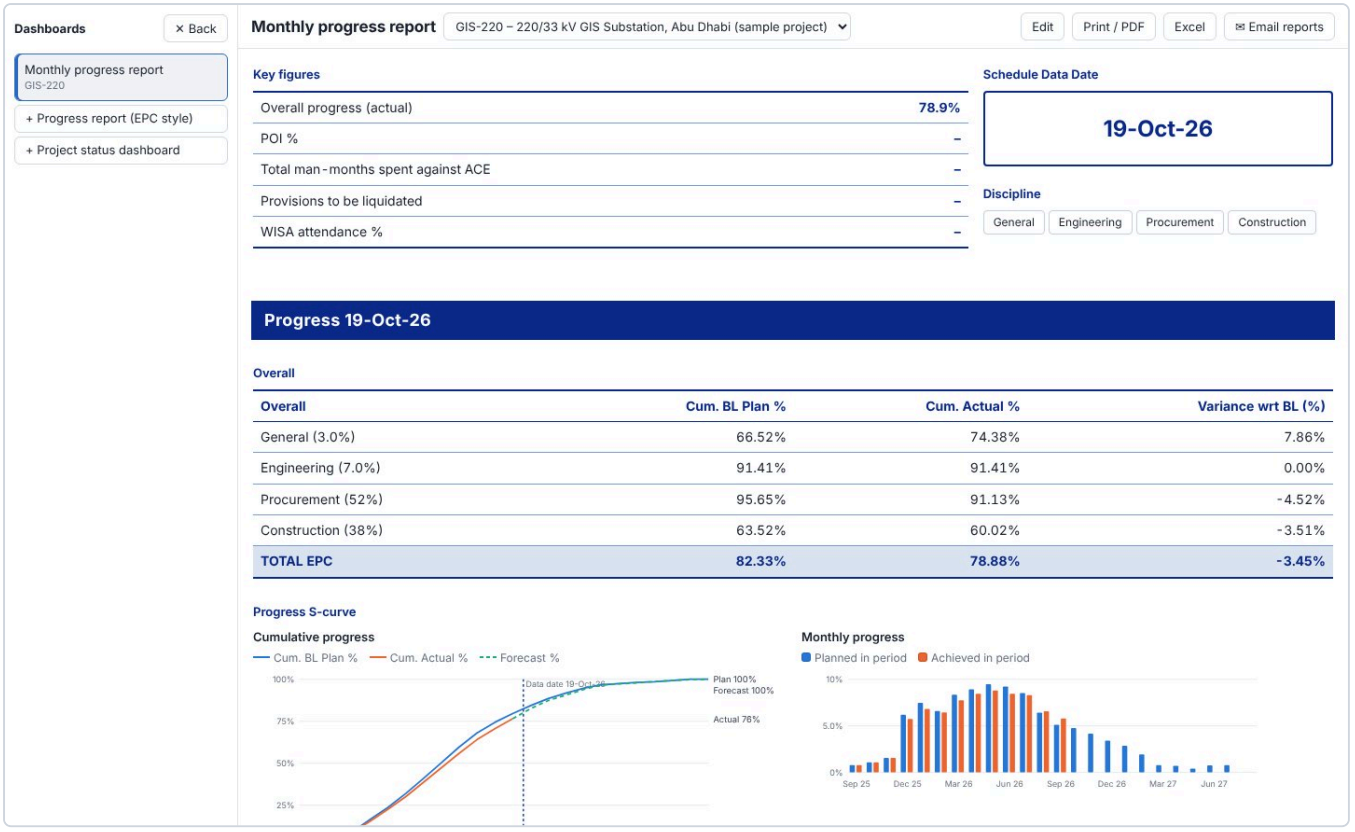


Figure 29. The Progress report (EPC style) template on the example project: key figures, the Schedule Data Date card, the Discipline slicer, the navy Progress banner and the weighted progress table with its TOTAL EPC row.

Progress is weighted by budgeted cost, labour hours, duration or equally, and grouped by WBS level 1 or 2, any activity code or any user field. With the example's cost loading, the weights come out at:

WBS level 1	Weight by cost	Cum. BL plan %	Cum. actual %	Variance
General	3.0%	66.52%	74.38%	7.86%
Engineering	7.0%	91.41%	91.41%	0.00%
Procurement	52.1%	95.65%	91.13%	-4.52%
Construction	37.8%	63.52%	60.02%	-3.51%
TOTAL	100%	**82.33%**	**78.88%**	**-3.45%**

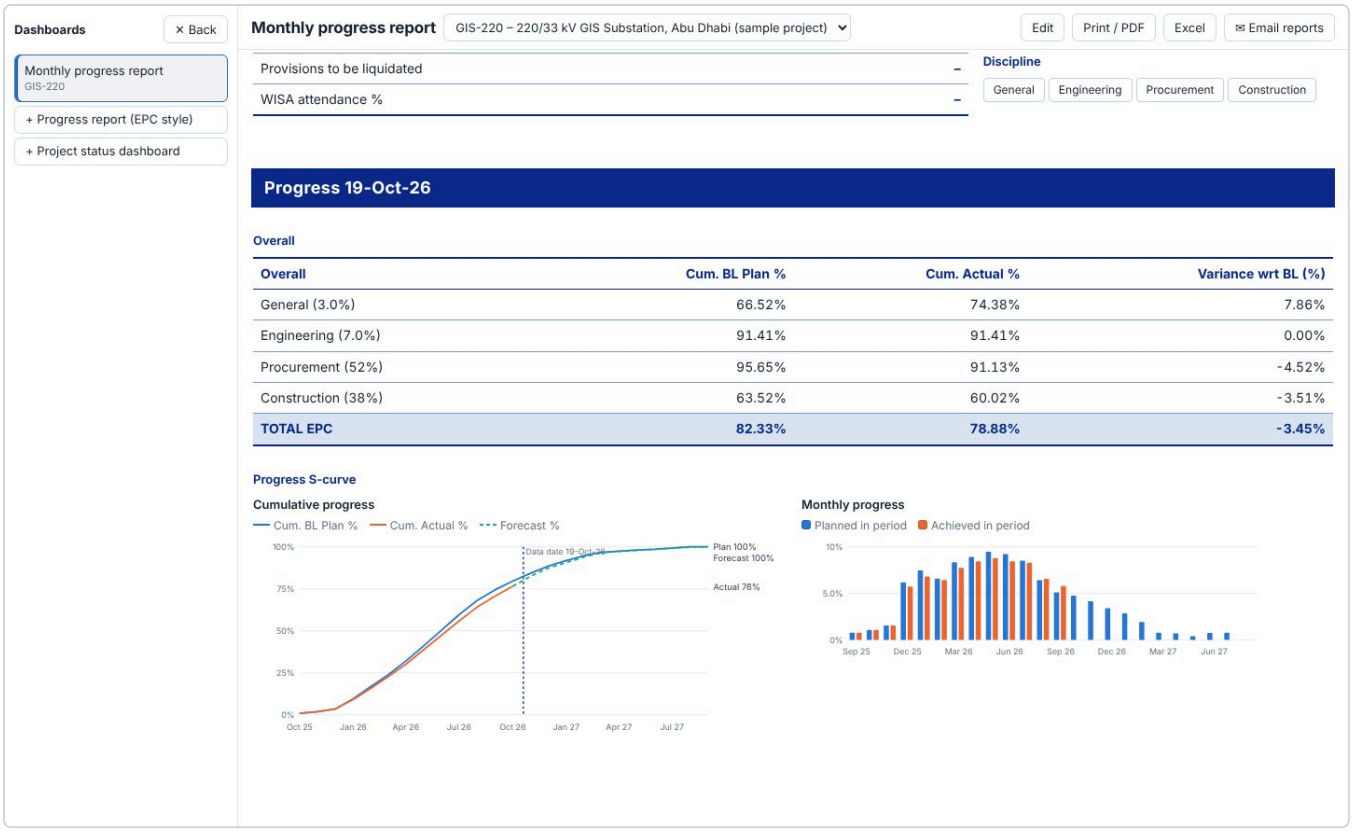


Figure 30. Lower half of the report: cumulative BL plan, actual and forecast S-curve, with monthly planned and achieved progress beside it.

Slicers

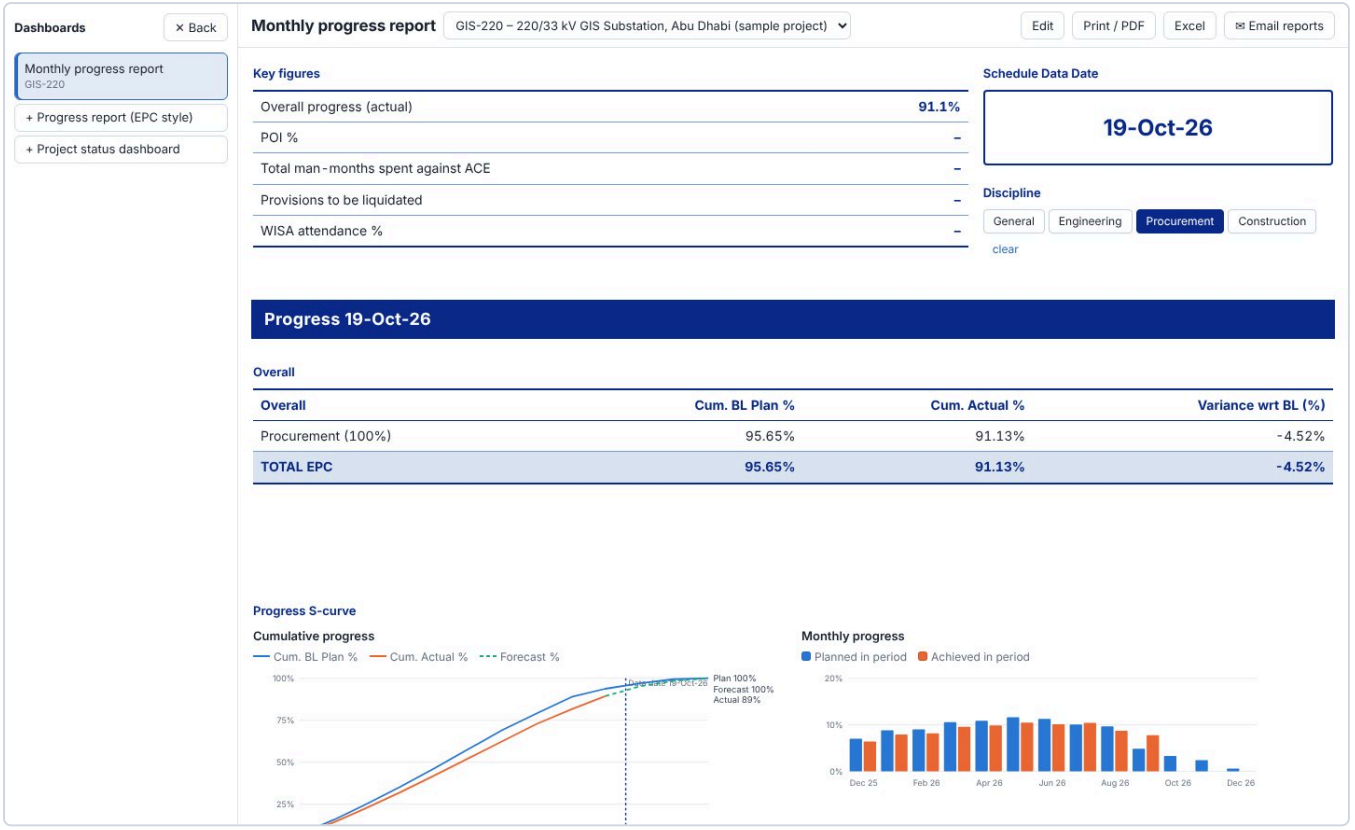


Figure 31. Clicking Procurement in the slicer filters every widget to procurement activities; the TOTAL row recomputes for the selection. Click again (or clear) to remove the filter.

Editing widgets and manual weightage

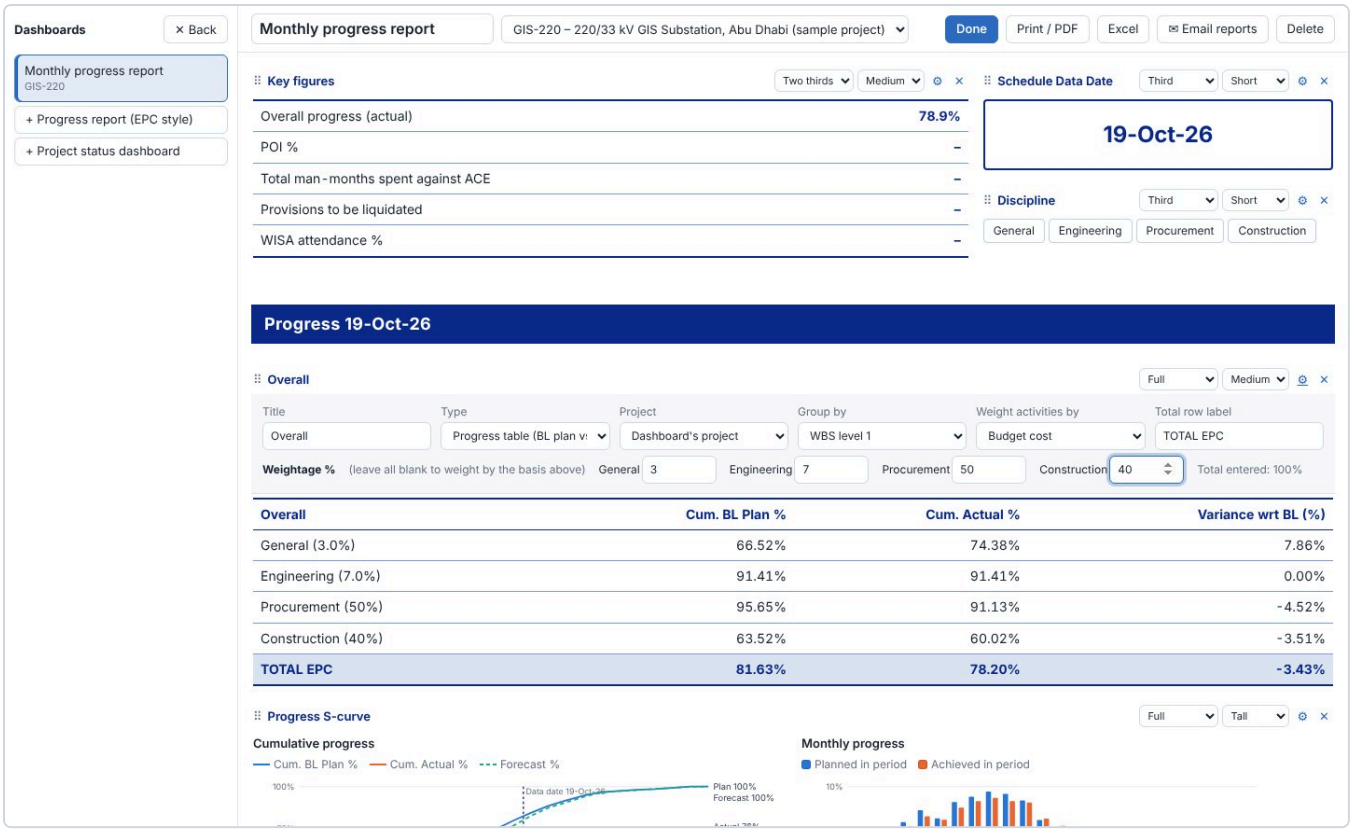


Figure 32. Edit mode. Each widget has width, height and a settings panel; here the progress table's Weightage % is typed in as 3 / 7 / 50 / 40.

Typing a weightage per group overrides the calculated weights, so a report can follow the contract's agreed split. With 3 / 7 / 50 / 40 the example reports a cumulative plan of 81.63%, actual 78.20% and variance -3.43%.

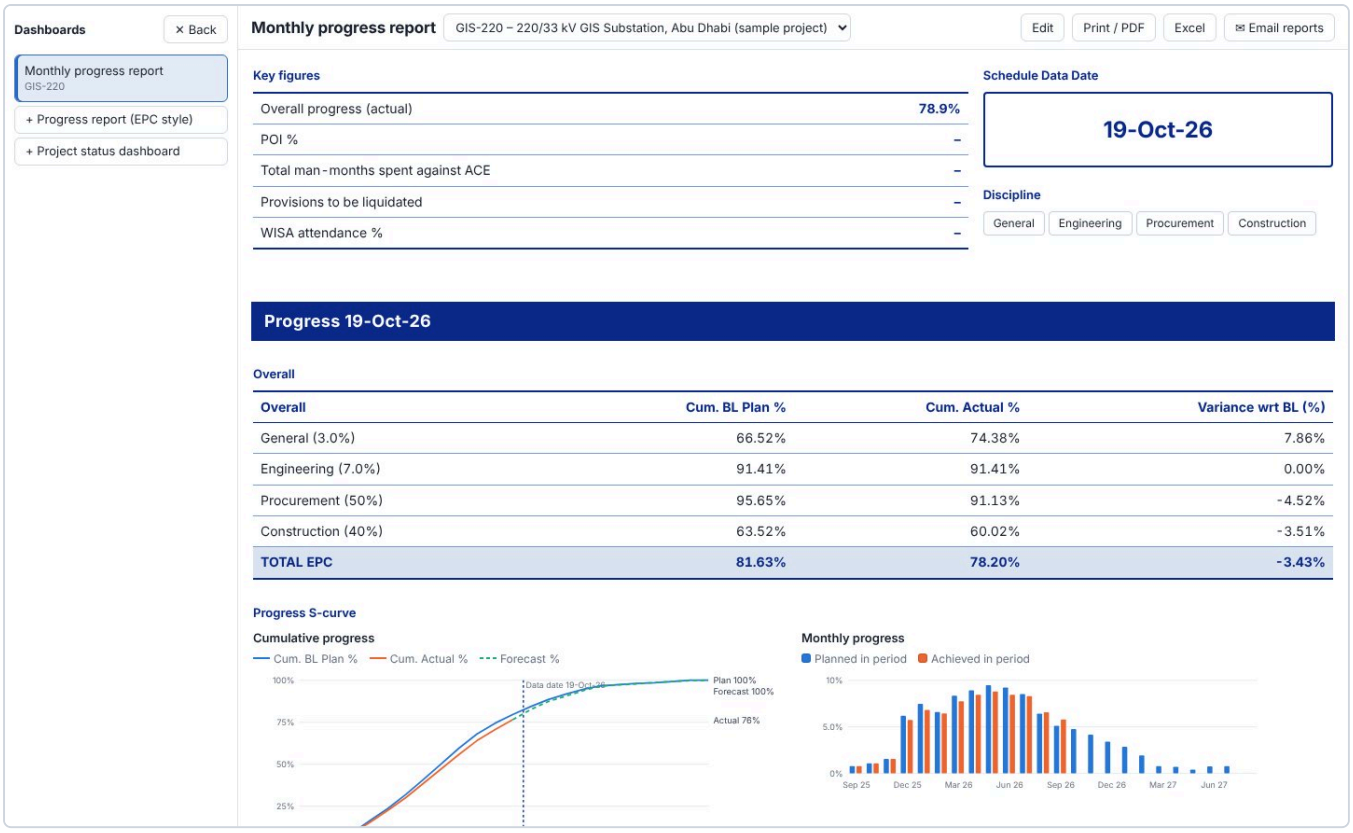


Figure 33. The report with the manual weightage: each row label shows its weight, and TOTAL EPC uses the typed split.

The key-figures table mixes calculated values (such as overall progress) with rows you type yourself, for figures that do not come from the schedule, for example POI %, man-months against ACE, provisions or WISA attendance.

Project status dashboard

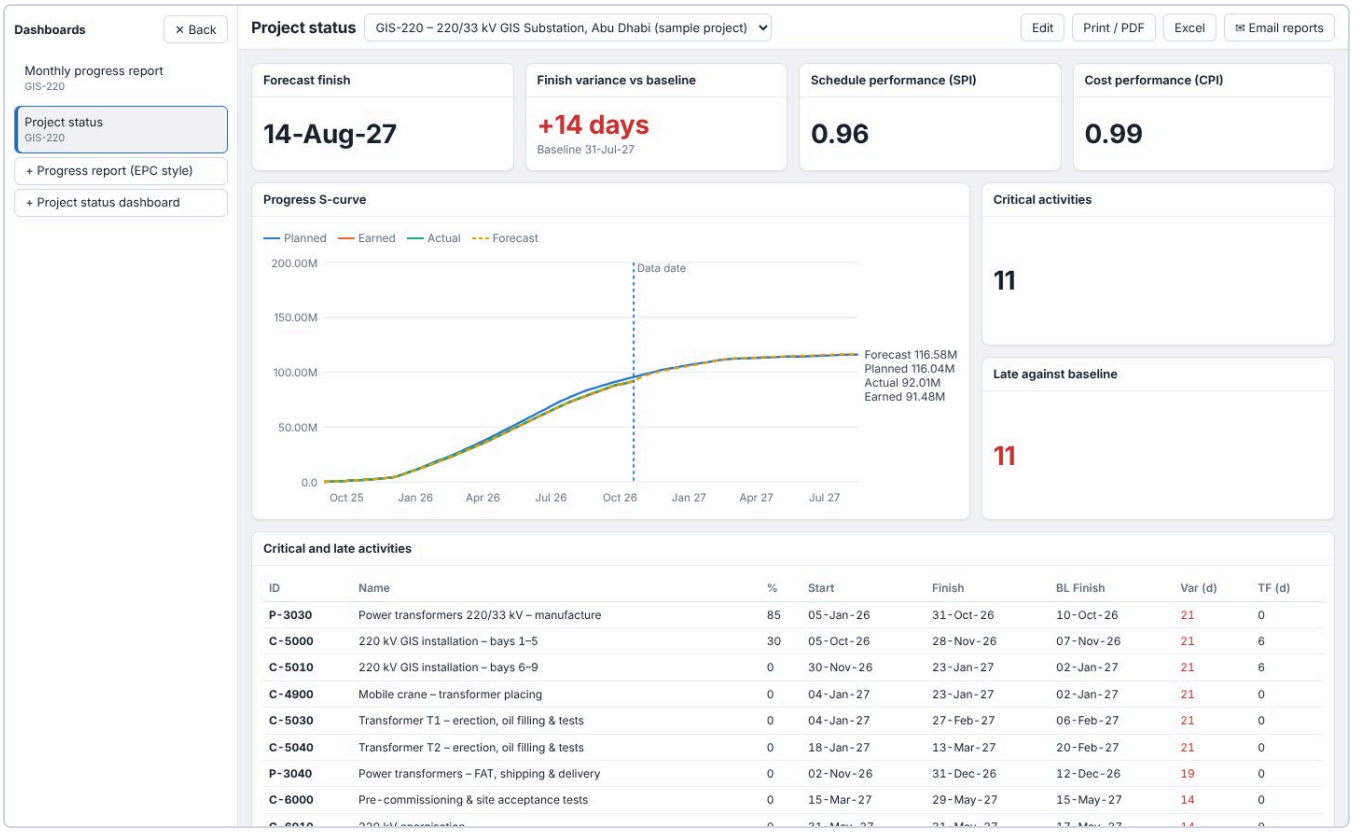


Figure 34. Project status template: forecast finish 14-Aug-27 (+14 days against the 31-Jul-27 baseline), SPI 0.96, CPI 0.99, the cost S-curve, critical and late counts, and the critical and late activities.

Available widgets: KPI tile, S-curve, resource histogram, activity table (critical, late, starting soon, look-ahead), portfolio table across all projects, quantity curve, text, progress table, progress S-curve, key-figures table, slicer and section banner.

PDF and Excel

Print / PDF prints the dashboard on A4 landscape with a header and footer; reports always print in the light theme. *Excel* builds a workbook: summary, progress table, progress S-curve, late and critical, three-week look-ahead, all activities, S-curve data, weekly resources and portfolio.

Emailed reports

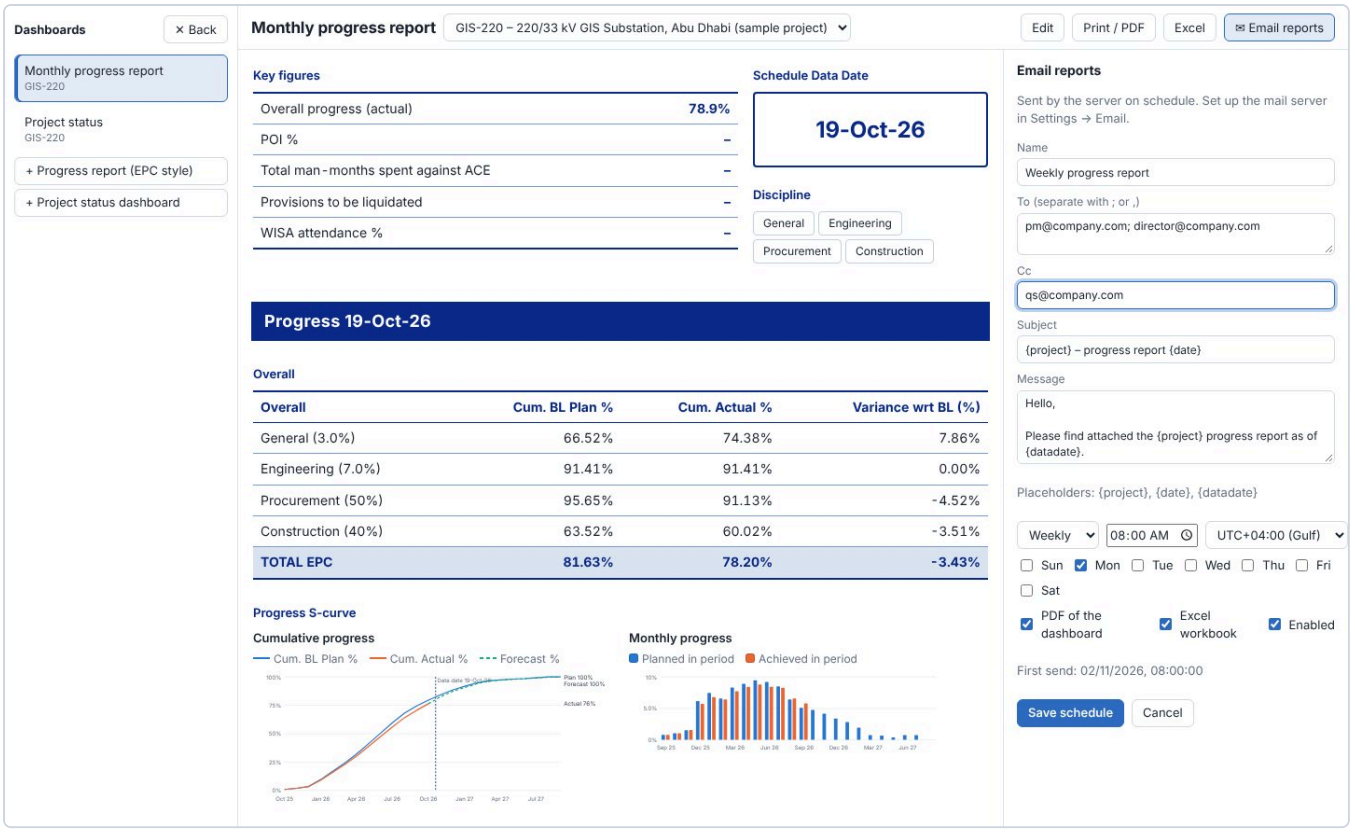


Figure 35. Email reports panel: a weekly schedule sending the report as PDF and Excel every Monday at 08:00 Gulf time to the project manager and director, copied to the QS.

Schedules run daily, weekly (on chosen days) or monthly (on a day of the month, or the last day). Subjects and messages accept `{project}`, `{date}` and `{datadate}`. Each run is logged with its result, and *Send now* sends immediately. On Stratum Server schedules run around the clock; in the desktop app they run while the app is open.

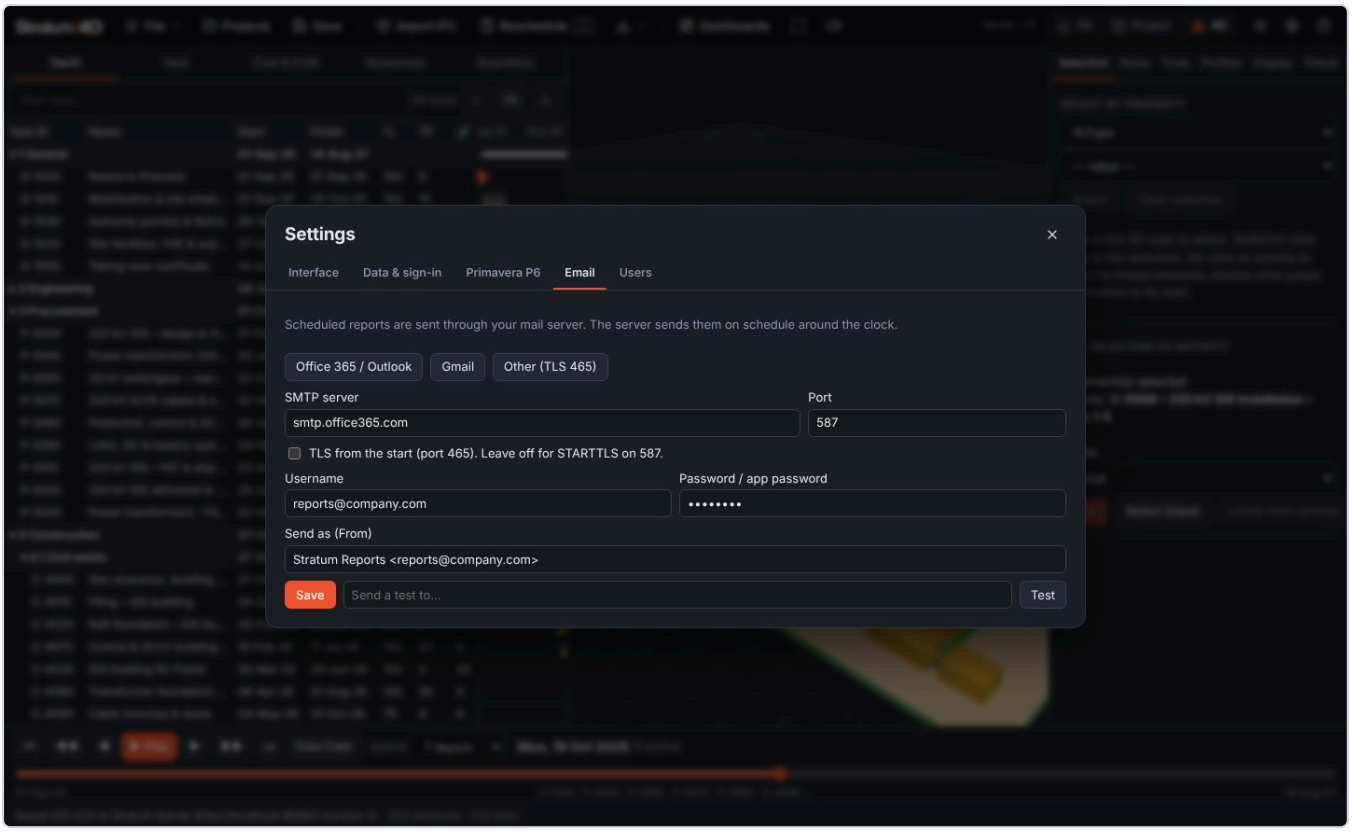


Figure 36. Settings → Email: the SMTP server (presets for Office 365 and Gmail), sender address and a test message.

11. Stratum Server for teams

Stratum Server is the shared version: one PostgreSQL database, sign-in, roles, P6 connections, scheduled emails and server-side PDF rendering. It ships as a Docker image with a Compose file that adds PostgreSQL 16.

```
cp .env.example .env      # set STRATUM_SECRET and POSTGRES_PASSWORD
docker compose up -d      # PostgreSQL 16 + Stratum Server on port 8080
```

Open `http://<server>:8080` and create the first administrator. In production, put the server behind your usual HTTPS front end (IIS, nginx or a load balancer).

Setting	Purpose
`STRATUM_SECRET`	Signs sign-in sessions and encrypts stored SMTP and P6 passwords. Keep it secret and do not change it later.
`DATABASE_URL`	PostgreSQL connection. Without it the server uses a SQLite file (`STRATUM_DB_FILE`).
`PORT` , `PUBLIC_URL`	Listening port (default 8080) and the address users see.
`NODE_EXTRA_CA_CERTS`	Company root certificate for reaching P6 over HTTPS.
`HTTPS_PROXY` , `NO_PROXY`	Outbound proxy for reaching P6.

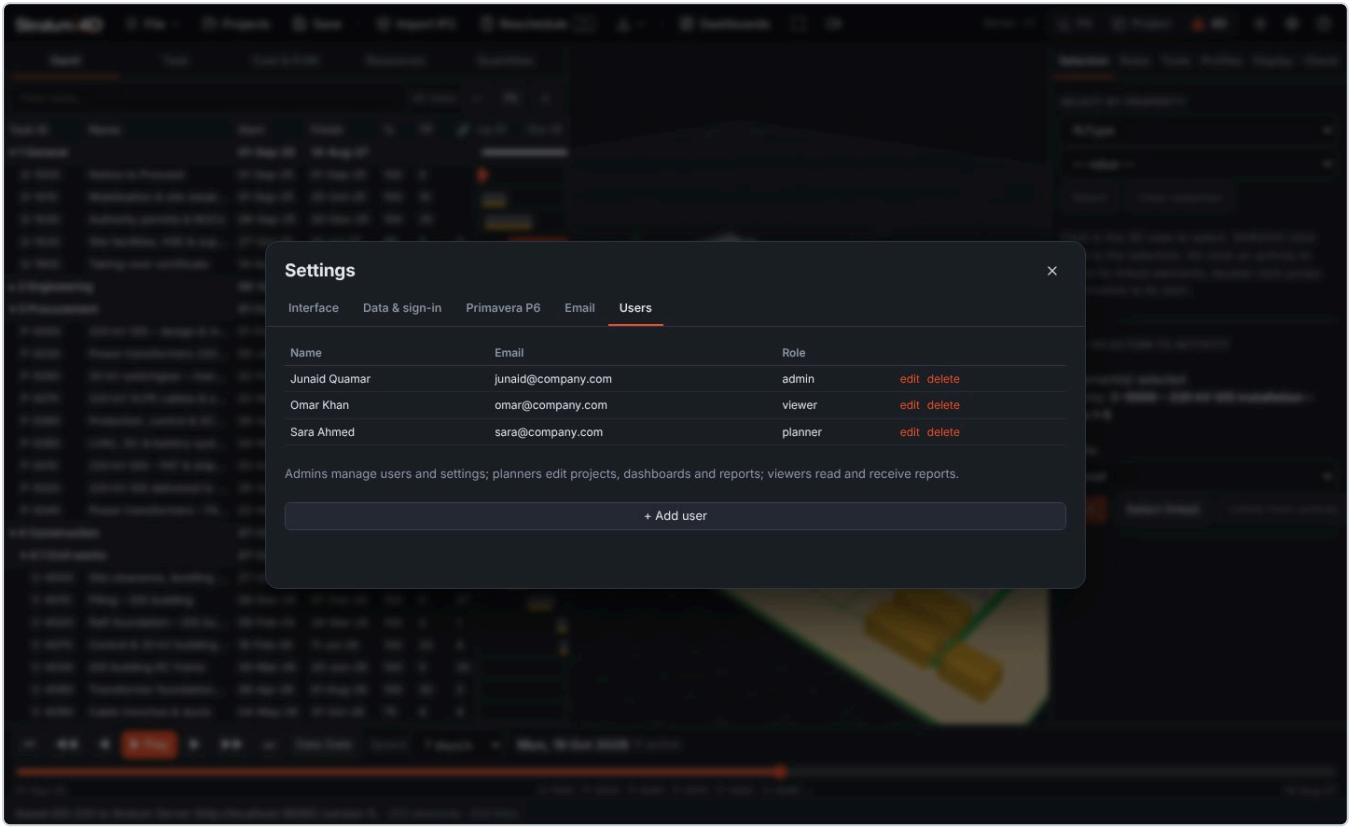


Figure 37. Settings → Users (administrators only): add users and give them a role.

Role	Can
Admin	Everything, including users, SMTP and P6 connection settings

Role	Can
Planner	Create and edit projects, dashboards and email schedules; import from P6
Viewer	Open projects and dashboards, and receive reports

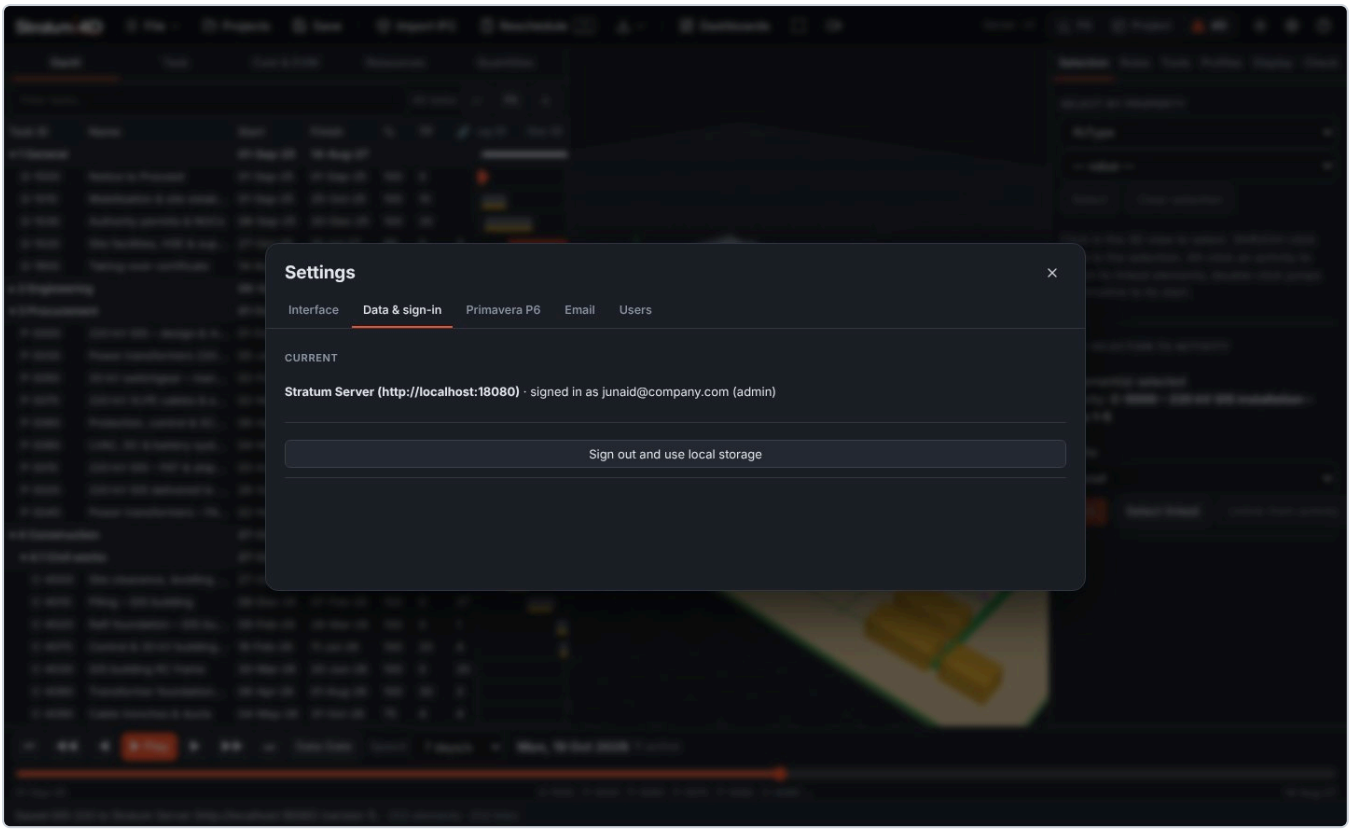


Figure 38. Settings → Data & sign-in: which database this client uses, and switching between local storage and a server.

12. Keyboard shortcuts

Keys	Action
`Space`	Play / pause
`←` `→`	Step one day
`Shift` + `←` `→`	Step one week
`Home` / `End`	Start / finish of the schedule
`F9`	Schedule (CPM)
`Ctrl` + `S`	Save to the database
`Ctrl` + `Alt` + `1` / `2` / `3`	P6 / MS Project / 4D layout
Click, `Shift`-click in 3D	Select, add to selection
`Alt`-click an activity	Select its linked elements
Double-click an activity	Move the timeline to its start
`Esc`	Cancel path drawing, close dialogs
`?`	Getting-started help

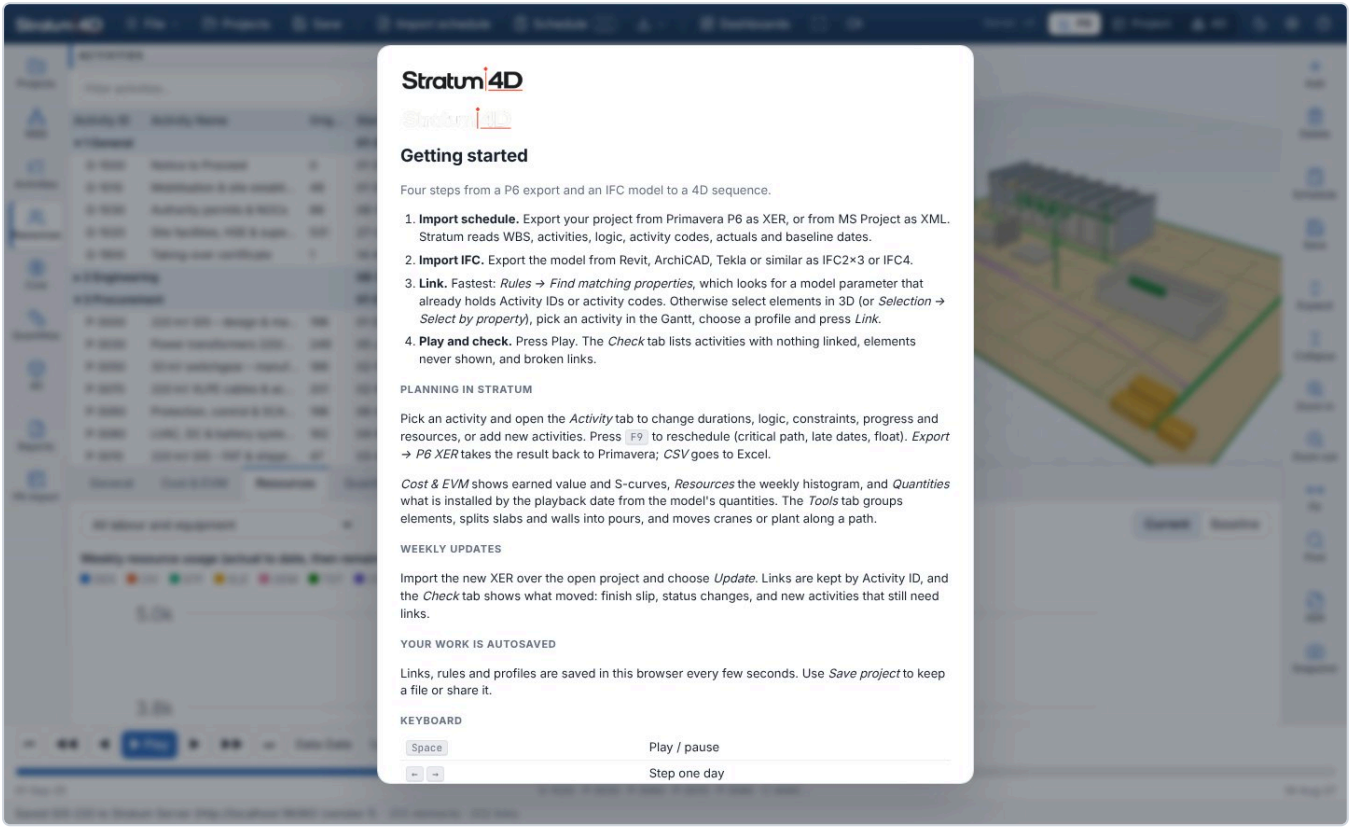


Figure 39. The getting-started help (?): four steps from a P6 export and an IFC model to a 4D sequence, plus shortcuts.

13. Troubleshooting

Symptom	Likely cause	What to do
P6 test stops at *Server name*	Not on the company network, or a typo	Connect to the VPN; check the address
P6 test stops at *Port open* with no answer	Firewall between this machine and P6	Ask IT to allow this computer (or the Stratum Server) to reach the P6 port
P6 test stops at *Certificate*	Company certificate authority not trusted	Install the root certificate in Windows/macOS, paste it into the connection, or set NODE_EXTRA_CA_CERTS on the server
P6 test stops at *P6 REST API* (404)	REST API not deployed at that path	Check the address ends in /p6ws/restapi; ask the P6 admin
Sign-in 401	Wrong user, password or database name	Check in the P6 web client; use the database name from the P6 login page
Sign-in 403	User has no API access	P6 Admin → Users → module access for web services / REST
No projects visible	No OBS access	Give the user project access in P6
“Not saved: someone else saved this project”	Another user saved a newer version	Use Save as new, or reopen the project and redo the change
Links lost after opening a project	The model is not stored in the database	Open the IFC again; links match by GlobalId
Emails not sent	SMTP settings, or the desktop app was closed	Send a test from Settings → Email; use Stratum Server for round-the-clock schedules
Windows protected your PC	Installer not code-signed yet	More info → Run anyway

14. Known limits

- The CPM engine follows P6's default options. Progress override, resource levelling and multiple float paths are not implemented yet.
- Costs and quantities are spread linearly over each activity, so weekly figures can differ slightly from P6's calendar-weighted spread.
- XER export has been checked by re-importing into Stratum, not yet against a live P6 database. Use P6's import preview the first time.
- The P6 EPPM connector has been tested against a mock of the documented REST API; the P6 database reader against SQL Server 2022 and SQLite copies of the P6 schema. Oracle has not been run against a live instance yet.
- Very large IFC files load slowly; GPU instancing and level of detail are planned.
- Undo/redo, column chooser, group and sort, saved filters and in-app baseline management are on the roadmap.
- Video export is WebM only.

Appendix A · Example project activities

All 46 activities of the example as imported (data date 19-Oct-26). OD = original duration in working days; TF = total float in days; ● = critical after F9. Baseline finish is shown only where it differs.

ID	Activity	WBS	OD	Start	Finish	BL finish	%	TF	Crit.
G-1000	Notice to Proceed	1	0	01-Sep-25	01-Sep-25		100	0	
G-1010	Mobilisation & site establishment	1	48	01-Sep-25	25-Oct-25		100	10	
G-1020	Site facilities, HSE & supervision	1	531	27-Oct-25	31-Jul-27		66	0	●
G-1030	Authority permits & NOCs	1	88	08-Sep-25	20-Dec-25		100	20	
G-1900	Taking-over certificate	1	1	14-Aug-27	14-Aug-27	31-Jul-27	0	0	●
E-2000	Basic design & single line diagram	2	60	08-Sep-25	15-Nov-25		100	0	
E-2010	Geotechnical & topographic survey	2	36	08-Sep-25	18-Oct-25		100	15	
E-2020	Civil & structural design – GIS building	2	87	20-Oct-25	31-Jan-26		100	0	
E-2030	Primary design – layouts, earthing, cable sizing	2	99	17-Nov-25	14-Mar-26		100	6	
E-2040	Secondary design – protection, control & SCADA	2	212	15-Dec-25	29-Aug-26	25-Jul-26	100	12	
E-2050	Client approval of detailed design	2	37	16-Mar-26	30-Apr-26	15-Apr-26	100	0	
E-2060	As-built drawings & O&M manuals	2	53	01-Jun-27	31-Jul-27		0	0	
P-3000	220 kV GIS – design & manufacture	3	198	01-Dec-25	31-Jul-26	11-Jul-26	100	0	
P-3010	220 kV GIS – FAT & shipping	3	47	03-Aug-26	26-Sep-26	05-Sep-26	100	0	
P-3020	220 kV GIS delivered to site	3	0	28-Sep-26	28-Sep-26	07-Sep-26	100	0	
P-3030	Power transformers 220/33 kV – manufacture	3	249	05-Jan-26	31-Oct-26	10-Oct-26	85	0	●
P-3040	Power transformers – FAT, shipping & delivery	3	50	02-Nov-26	31-Dec-26	12-Dec-26	0	0	●
P-3050	33 kV switchgear – manufacture & delivery	3	189	02-Feb-26	19-Sep-26		100	18	
P-3060	Protection, control & SCADA panels	3	198	06-Apr-26	28-Nov-26		70	8	
P-3070	220 kV XLPE cables & accessories	3	201	02-Mar-26	31-Oct-26		90	6	
P-3080	LVAC, DC & battery systems	3	162	04-May-26	14-Nov-26		60	14	
C-4000	Site clearance, levelling & temporary fencing	4.4.1	34	27-Oct-25	06-Dec-25		100	0	
C-4010	Piling – GIS building	4.4.1	53	08-Dec-25	07-Feb-26	31-Jan-26	100	0	
C-4020	Raft foundation – GIS building	4.4.1	39	09-Feb-26	28-Mar-26		100	0	
C-4030	GIS building RC frame	4.4.1	67	30-Mar-26	20-Jun-26		100	0	
C-4040	GIS building roof steel & cladding	4.4.1	59	22-Jun-26	29-Aug-26	15-Aug-26	100	0	
C-4050	EOT crane installation & load test	4.4.1	18	31-Aug-26	19-Sep-26		100	0	
C-4060	GIS building finishes & HVAC	4.4.1	72	07-Sep-26	28-Nov-26		45	10	
C-4070	Control & 33 kV building – foundations & structure	4.4.1	118	16-Feb-26	11-Jul-26		100	20	
C-4080	Transformer foundations, oil pits & firewall	4.4.1	97	06-Apr-26	01-Aug-26		100	30	
C-4090	Cable trenches & ducts	4.4.1	150	04-May-26	31-Oct-26		75	4	
C-4100	Roads, drainage, boundary wall & gatehouse	4.4.1	141	14-Sep-26	27-Feb-27		15	40	
C-4900	Mobile crane – transformer placing	4.4.1	18	04-Jan-27	23-Jan-27	02-Jan-27	0	0	●
C-5000	220 kV GIS installation – bays 1–5	4.4.2	48	05-Oct-26	28-Nov-26	07-Nov-26	30	6	
C-5010	220 kV GIS installation – bays 6–9	4.4.2	45	30-Nov-26	23-Jan-27	02-Jan-27	0	6	

ID	Activity	WBS	OD	Start	Finish	BL finish	%	TF	Crit.
C-5020	220 kV GIS HV testing	4.4.2	24	25-Jan-27	20-Feb-27		0	6	
C-5030	Transformer T1 – erection, oil filling & tests	4.4.2	48	04-Jan-27	27-Feb-27	06-Feb-27	0	0	•
C-5040	Transformer T2 – erection, oil filling & tests	4.4.2	45	18-Jan-27	13-Mar-27	20-Feb-27	0	0	•
C-5050	220 kV cable laying & terminations	4.4.2	87	02-Nov-26	13-Feb-27		0	10	
C-5060	33 kV switchgear installation	4.4.2	46	12-Oct-26	05-Dec-26		10	12	
C-5070	Protection, control & SCADA panels installation	4.4.2	77	07-Dec-26	06-Mar-27		0	4	
C-5080	Earthing & lightning protection	4.4.2	214	06-Apr-26	19-Dec-26		70	30	
C-5090	LVAC, DC & battery installation	4.4.2	47	07-Dec-26	30-Jan-27		0	14	
C-6000	Pre-commissioning & site acceptance tests	4.4.3	63	15-Mar-27	29-May-27	15-May-27	0	0	•
C-6010	220 kV energisation	4.4.3	0	31-May-27	31-May-27	17-May-27	0	0	•
C-6020	Trial operation & handover	4.4.3	65	01-Jun-27	14-Aug-27	31-Jul-27	0	0	•

Appendix B · Terms in each layout

Concept	P6 style	MS Project style	4D style
Activity	Activity	Task	Task
Activities	Activities	Tasks	Tasks
Activity Id	Activity ID	ID	Task ID
Activity Name	Activity Name	Task Name	Task Name
Wbs	WBS	Outline	WBS
Data Date	Data Date	Status Date	Data Date
Total Float	Total Float	Total Slack	Total Float
Free Float	Free Float	Free Slack	Free Float
Original Duration	Original Duration	Duration	Planned Duration
Remaining Duration	Remaining Duration	Remaining Duration	Remaining Duration
Percent Complete	Physical % Complete	% Complete	% Complete
Predecessors	Predecessors	Predecessors	Predecessors
Successors	Successors	Successors	Successors
Relationship	Relationship	Link	Dependency
Baseline	Baseline	Baseline	Baseline
Schedule	Schedule	Calculate	Reschedule
Resources	Resources	Resource Names	Resources
Constraint	Primary Constraint	Constraint Type	Constraint
Milestone	Milestone	Milestone	Milestone
Project	Project	Project	Project
Projects	Projects	Projects	Projects
Playback Date	Timescale date	Date	Focus date